

Prediction Market Audit & Threat Modeling

김상현

2026.04.08

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Project Background

1.1 Project Motivation

What is Prediction Market

미래에 객관적으로 검증할 수 있는 모든 사건이나 결과가 시장이 될 수 있음
(etc: 선거 결과, 퇴임 시점, 금리 인상/인하, 종전 시점 등)



Elections · Trump

Trump out as President before 2027?

Past ▾

Mar 31

Apr 30

Jun 30

Dec 31

17% chance ▲ 4%

Polymarket



\$5,449,356 Vol. | ⌚ Dec 31, 2026

1H 6H 1D 1W 1M ALL 🔍 ⚙️

Buy Sell

Market ▾

Yes 17¢

No 84¢

Amount

\$0

+\$1

+\$5

+\$10

+\$100

Max

Trade

By trading, you agree to the [Terms of Use](#).

All

Politics

Trump

Mentions



Will Trump resign by December 31, 2026?

7%

What is Prediction Market

미래에 객관적으로 검증할 수 있는 모든 사건이나 결과가 시장이 될 수 있음
(etc: 선거 결과, 퇴임 시점, 금리 인상/인하, 종전 시점 등)



Economy · Economic Policy

Fed decision in April?



Past ▾

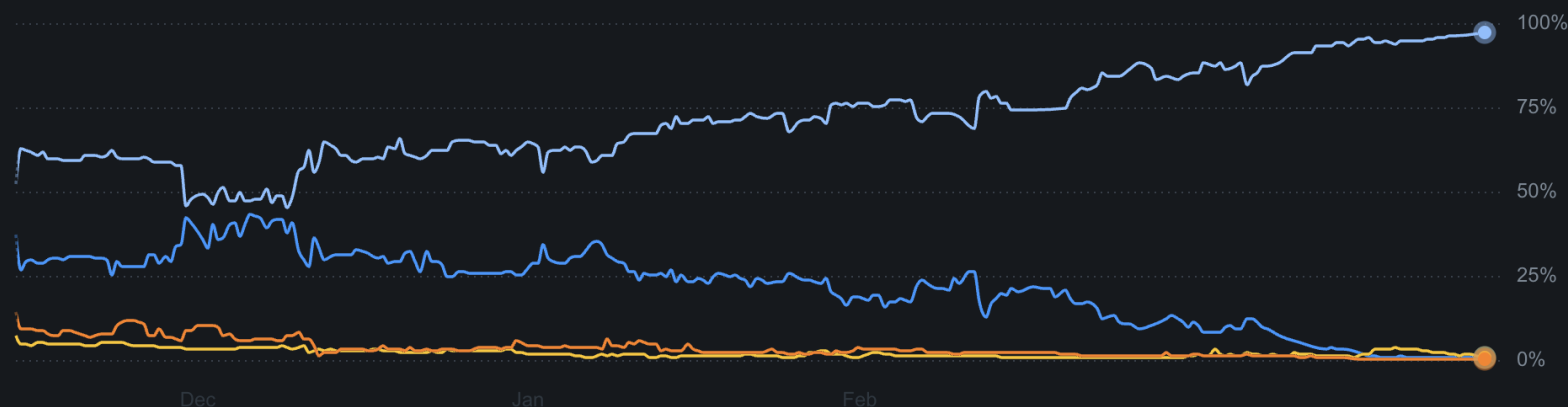
Apr 29

Jun 17

Jul 29

● No change 97.7% ● 25 bps decrease 1.1% ● 25+ bps increase <1% ● 50+ bps decrease <1%

Polymarket



\$40,262,428 Vol. | Apr 29, 2026

1H 6H 1D 1W 1M ALL ▾ ⚙



50+ bps decrease

Buy Sell

Market ▾

Yes 0.5¢

No 99.6¢

Amount

\$0

+\$1

+\$5

+\$10

+\$100

Max

Trade

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What is Prediction Market

미래에 객관적으로 검증할 수 있는 모든 사건이나 결과가 시장이 될 수 있음
(etc: 선거 결과, 퇴임 시점, 금리 인상/인하, 종전 시점 등)



Geopolitics · Trump

US x Iran ceasefire by...?



Polymarket

\$65,857,368 Vol.

March 31

\$41,354,482 Vol. 

1%

Buy Yes 0.6¢

Buy No 99.5¢

April 7

\$1,856,723 Vol. 

9%

Buy Yes 9¢

Buy No 92¢

April 15

\$6,453,574 Vol. 

20%

Buy Yes 20¢

Buy No 81¢

April 30

\$7,136,864 Vol. 

34% ▼ 44%

Buy Yes 34¢

Buy No 67¢

May 31

\$2,437,176 Vol. 

51% ▼ 37%

Buy Yes 51¢

Buy No 50¢



March 31

Buy Sell

Market ▼

Yes 0.6¢

No 99.5¢

Amount

\$0

+\$1

+\$5

+\$10

+\$100

Max

Trade

By trading, you agree to the [Terms of Use](#).

What is Prediction Market

사람들은 금전적 리스크가 있을때 더 정확한 정보를 제공함



What is Prediction Market

사람들은 금전적 리스크가 있을때 더 정확한 정보를 제공함



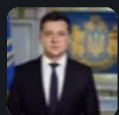
What is Prediction Market

판정 규칙의 모호성으로 인한 마켓 정산 분쟁 사례



What is Prediction Market

판정 규칙의 모호성으로 인한 마켓 정산 분쟁 사례

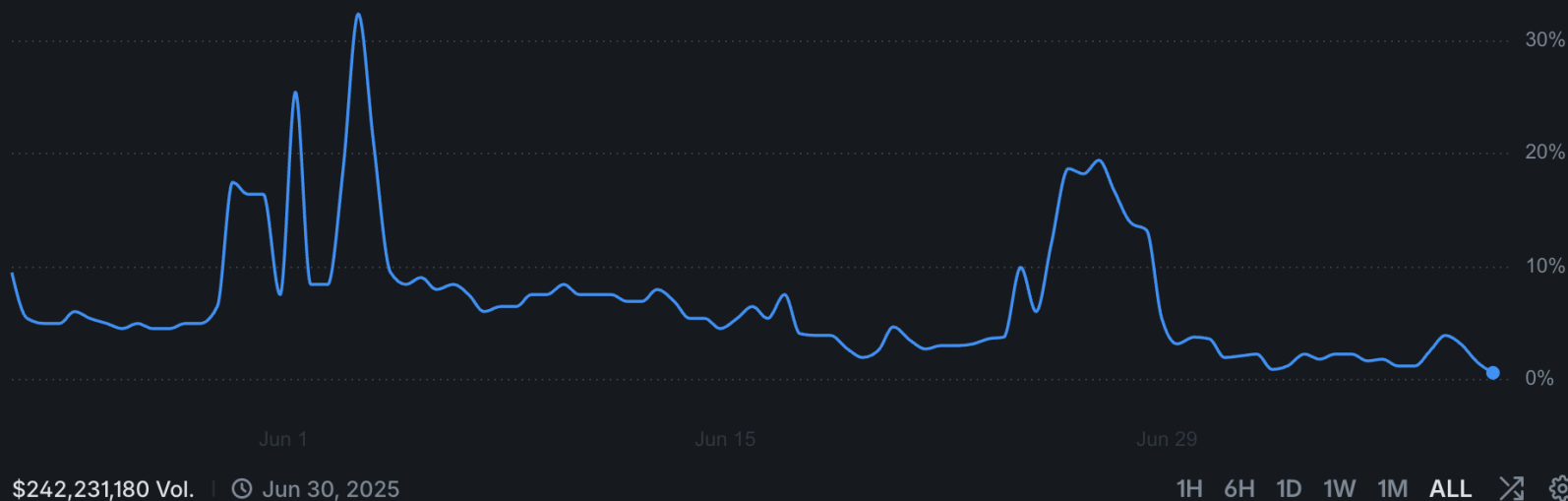


Will Zelenskyy wear a suit before July?

</> 🔗 📌

<1% chance ▼ 9%

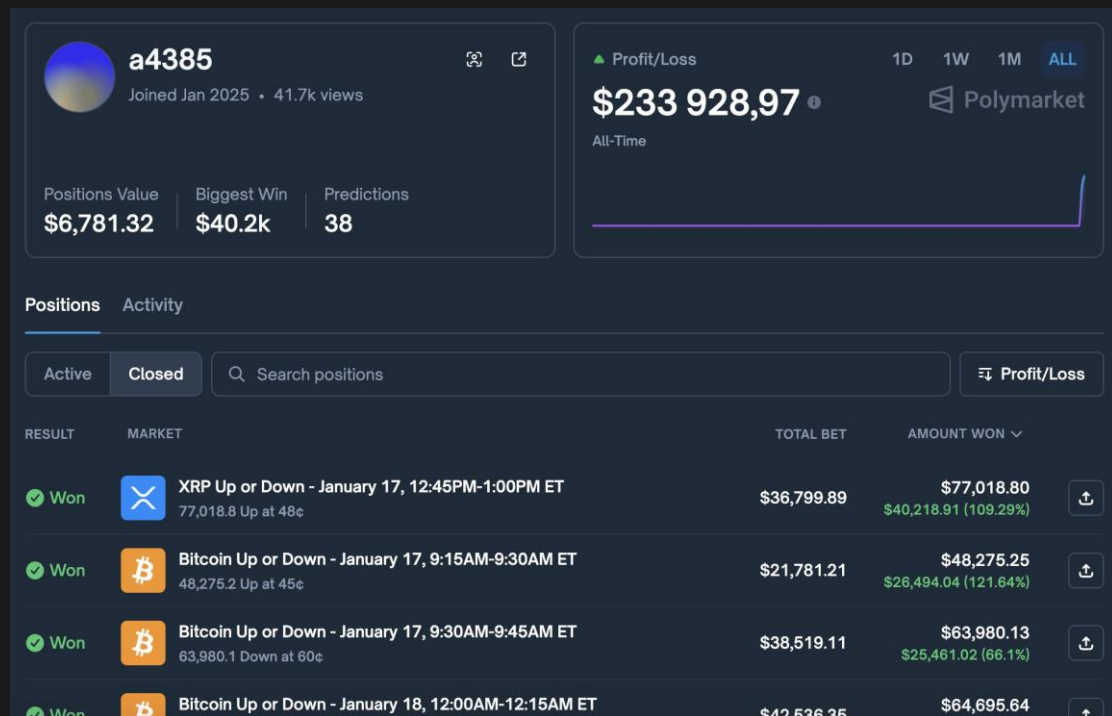
 Polymarket



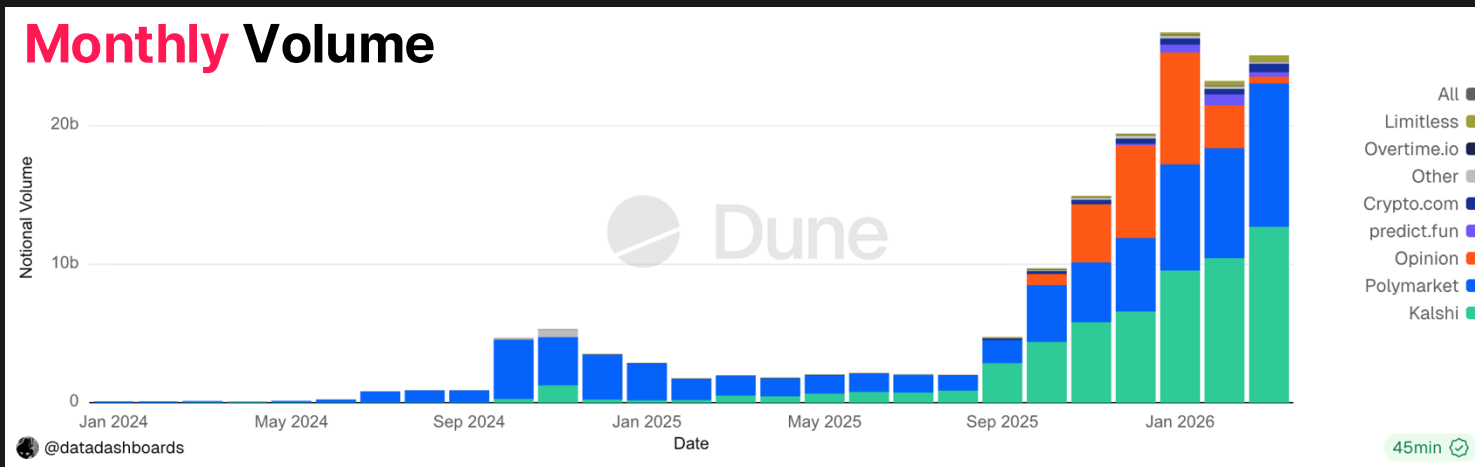
Outcome: No

What is Prediction Market

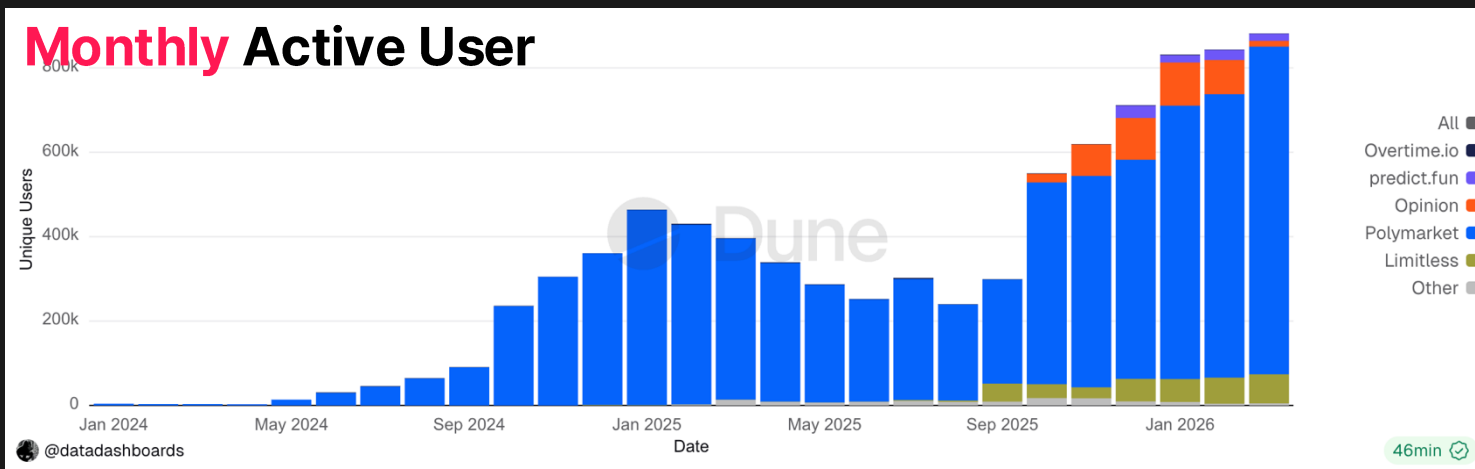
현물 유동성 부족을 악용한 가격 조작 사례



Prediction Market **Usage Is Exploding**



지난달 Volume
250억 달러



지난달 활성 사용자
88만 명

Why Prediction Markets **Need Threat Modeling?**

시장은 급성장하고 있지만, 이를 보호할 보안표준과 구조적 보조장치는 미비한 상황



표준·Best Practice 부재



프로토콜마다 다른 구현



반복되는 구조적 취약점

공통 리스크 정리와 프로토콜 간 비교가 어려워 **반복되는 취약점의 근본 원인 식별 불가**

Project Background

1.2 Core Components

Core Components

Market Configuration

- Market Creator
- Market Types
- Resolution Criteria

Trading Mechanism

- CLOB
- AMM

Oracle & Dispute

- Price Oracle
- Optimistic Oracle

Core Components

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Market Configuration

시장의 질문, 규칙, 기간, 정산 기준 등 기본 구성을 정의하는 단계

Review Market with  **OPINION AI**



US recession in 2026?
Created by User_47C52F with Opinion AI

YES
NO

End Date: 01/01/2027
Category:
Selected Trading Token:  USDT

Rules:

This market resolves to 'Yes' if, by the resolution time, the National Bureau of Economic Research (NBER) has officially declared that a US recession began at any point during the calendar year 2026. The declaration must be publicly available and explicitly state that a recession started in 2026 (even if the recession continues past 2026 or ends after 2026). If no such declaration is made by NBER by resolution time, or if the NBER has not issued a determination regarding any recession start in 2026, the market resolves to 'No'. If there is ambiguity or multiple conflicting interpretations, the most recent official statement from NBER published before the resolution time prevails.

 This market will be resolved by Opinion AI.

Back
Confirm & Create



US recession in 2026?
Created by User_47C52F with Opinion AI

YES
NO

End Date: 01/01/2027
Category:
Selected Trading Token:  USDT

YES/NO 선택지, 종료 날짜

Market Configuration

시장의 질문, 규칙, 기간, 정산 기준 등 기본 구성을 정의하는 단계

Review Market with  **OPINION AI**



US recession in 2026?
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YES
NO

End Date: 01/01/2027
Category:
Selected Trading Token:  USDT

Rules:

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Confirm & Create

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결과 판정 기준을 결정하는 기준에 대한 룰북

Core Components

Market Configuration

- Market Creator
- Market Types
- Resolution Criteria

Trading Mechanism

- CLOB
- AMM

Oracle & Dispute

- Price Oracle
- Optimistic Oracle

Trading Mechanism

참여자들이 시장 가격을 형성하며 예측 토큰을 사고파는 거래 단계

Order Book

Graph

Resolution

+ Rewards

TRADE YES

	PRICE	SHARES	TOTAL
	91¢	21,473.87	\$42,789.76
	90¢	23,902.01	\$23,248.54
	89¢	1,793.09	\$1,736.73
Asks	88¢	160.09	\$140.88

Last: 88¢

Spread: 1¢

Bids	87¢	1,607.37	\$1,398.41
	86¢	2,653.26	\$3,680.21
	85¢	1,445.56	\$4,908.94
	84¢	1,610.00	\$6,261.34

Central Limit Order Book(CLOB)


December 31

Buy
Sell
Limit ▼

Yes 88¢
No 13¢

Limit Price
- 88¢ +

Shares

-100
-10
+10
+100
+200

Set Expiration
☐

Total
\$0

To win ⓘ
\$0

Trade

지정가 구매창

Trading Mechanism

참여자들이 시장 가격을 형성하며 예측 토큰을 사고파는 거래 단계

Outcomes



#1 No

[Add Liquidity](#)
[View pool details](#)
[New conditional market](#)

99.7%

☒



#2 Yes

[Add Liquidity](#)
[View pool details](#)
[New conditional market](#)

0.3%

☐



#3 Invalid ?

[Add Liquidity](#)
[New conditional market](#)

NA ?

☐

Comments

Related Conditional Markets

Top Holders

Activity

Leave a comment



No

Market ▾

You pay

0



SEER_CREDITS ▾

0 SEER_CREDITS Max

↓

You will get

0



NO_JESUS

Avg price

Potential return ?

Parameters: Max slippage 1% - Instant ⚙

0.000 SEER_CREDITS

0.000 SEER_CREDITS (0.00%)

AMM 구매창

Core Components

Market Configuration

- Market Creator
- Market Types
- Resolution Criteria

Trading Mechanism

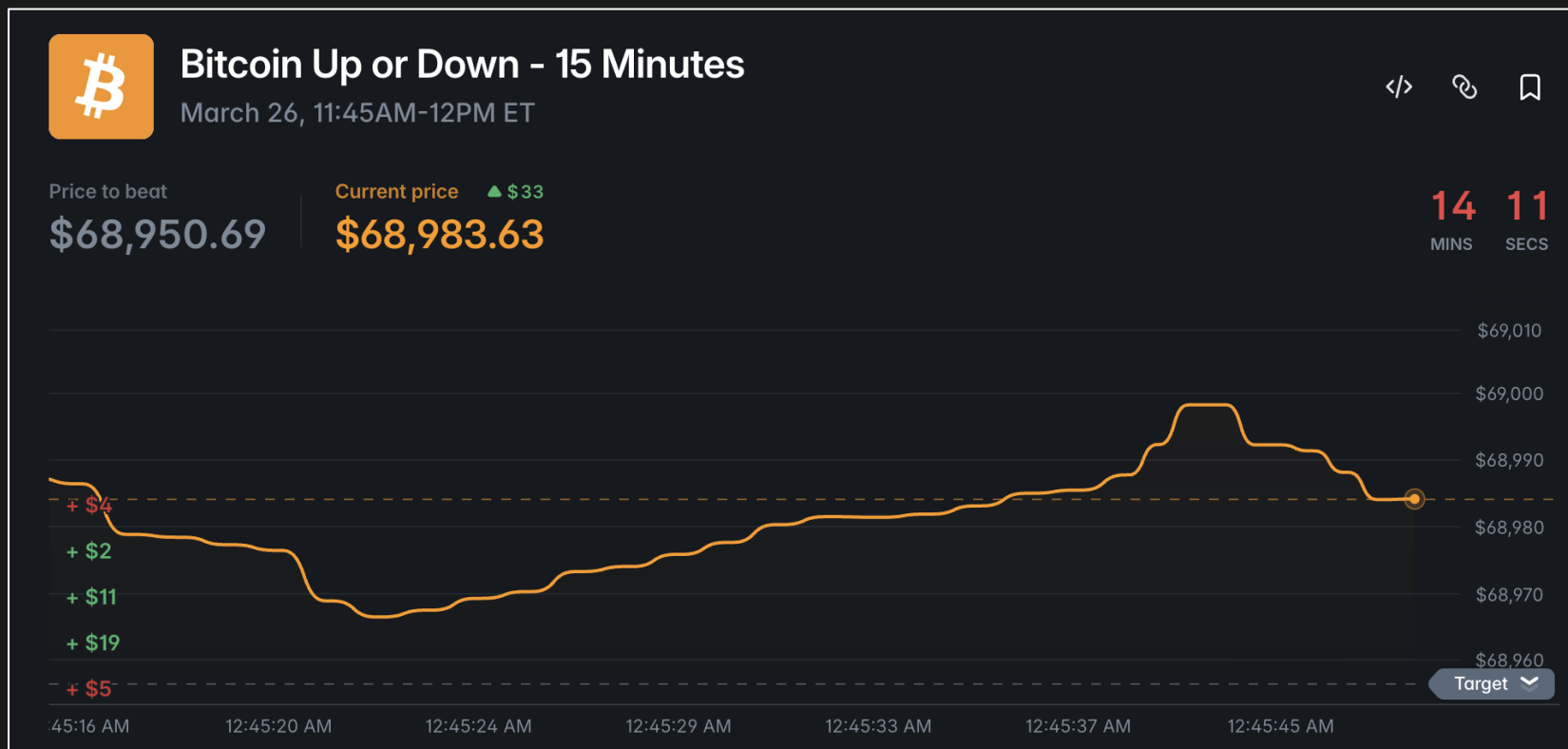
- CLOB
- AMM

Oracle & Dispute

- Price Oracle
- Optimistic Oracle

Oracle & Dispute

마켓이 종료되고 결과에 대한 데이터를 가져오는 단계



Crypto 시장

Oracle & Dispute

마켓이 종료되고 결과에 대한 데이터를 가져오는 단계

 Geopolitics · Trump		US x Iran ceasefire by...?		</> 🔗 📌	
\$52,489,199 Vol.				Polymarket	
March 31		5%		Buy Yes 5¢	Buy No 96¢
\$32,950,875 Vol. 📈					
April 7		15%		Buy Yes 15¢	Buy No 86¢
\$583,737 Vol. 📈					
April 15		28%		Buy Yes 28¢	Buy No 73¢
\$5,006,557 Vol. 📈					
April 30		41% ▼ 35%		Buy Yes 41¢	Buy No 60¢
\$5,771,055 Vol. 📈					
May 31		56% ▼ 31%		Buy Yes 56¢	Buy No 45¢
\$2,068,953 Vol. 📈					



Optimistic Oracle

비정형화된 데이터 필요

Oracle & Dispute

마켓이 종료되고 결과에 대한 데이터를 가져오는 단계

Bitcoin Up or Down on December 7?

Propose Answer

Select option

Bond

USDC.e 750

Reward

USDC.e 2

Challenge period ends

Connect wallet

Polygon

Managed Optimistic Oracle V2

Event-based Expiry

Timestamp

Requested Time

UTC

Fri, 05 Dec 2025 17:00:07 GMT

UNIX

1764954007

Additional Text Data

Description

This market will resolve to "Up" if the "Close" price for the Binance 1 minute candle for BTC/USDT Dec 6 '25 12:00 in the ET timezone (noon) is lower than the final "Close" price for the Dec 7 '25 12:00 ET candle.

This market will resolve to "Down" if the "Close" price for the Binance 1 minute candle for BTC/USDT Dec 6 '25 12:00 in the ET timezone (noon) is higher than the final "Close" price for the Dec 7 '25 12:00 ET candle.

If the final "Close" price for both of these candles is exactly equal on Binance, this market will resolve 50-50.

The resolution source for this market is Binance, specifically the BTC/USDT "Close" prices currently available at https://www.binance.com/en/trade/BTC_USDT with "1m" and "Candles" selected on the top bar.

Select option

Down

(p1)

Up

(p2)

unknown/50-50

(p3)

Custom

제안할 의견 선택

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Oracle & Dispute

마켓이 종료되고 결과에 대한 데이터를 가져오는 단계

Bitcoin Up or Down on December 7?

Propose Answer

Select option

Bond ⓘ

USDC.e ⓘ 750

Reward ⓘ

USDC.e ⓘ 2

Challenge period ends ⓘ

Connect wallet

Polygon

Managed Optimistic Oracle V2

Event-based Expiry

Timestamp

Requested Time

UTC

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Bond ⓘ

USDC.e ⓘ 750

Reward ⓘ

USDC.e ⓘ 2

Challenge period ends ⓘ

보증금 필요

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Oracle & Dispute

마켓이 종료되고 결과에 대한 데이터를 가져오는 단계


ORACLE

[Verify](#)
[Propose](#)
[Settled](#)
[Docs ↗](#)


0xe8...C52F



Verify 355 statements

1

Proposers post a bond to assert that a piece of data is correct.

2

During the challenge period, data proposals are verified and can be disputed.

3

If correctly disputed, the data is not used and the challenger receives a reward.

Project

Chain

Oracle

Polymarket

Clear filters

Query	Proposal	Bond	Minimum challenge period left
Coinbase (COIN) Up or Down on March 26? Polymarket 03/25/2026, 9:01 PM Polygon Event-based	Down	750	1 m 5 s
Oilers vs. Golden Knights: O/U 6.5 Polymarket 02/28/2026, 1:31 AM Polygon Event-based	Over	750	1 m 17 s
Spread: Golden Knights (-1.5) Polymarket 03/26/2026, 1:21 AM Polygon Event-based	Oilers	750	1 m 27 s
Oilers vs. Golden Knights: O/U 7.5 Polymarket 02/28/2026, 1:32 AM Polygon Event-based	Under	750	1 m 27 s

Minimum challenge period left

1 m 5 s

1 m 17 s

1 m 27 s

1 m 27 s

Dispute 가능한 Challenge period 설정

Oracle & Dispute

마켓이 종료되고 결과에 대한 데이터를 가져오는 단계

 Will the highest temperature in Dallas be 56°F or higher on December 15?

 Dispute Proposal?

Proposal: No

Bond ⓘ USDC.e 750
 Reward ⓘ USDC.e 5
 Challenge period ends ⓘ 12/16/2025, 5:55 PM

Connect wallet

 Polygon

 Optimistic Oracle V2

 Event-based Expiry

 Timestamp

Requested Time
 UTC Sun, 14 Dec 2025 16:35:45 GMT
 UNIX 1765730145

Proposed Time
 UTC Tue, 16 Dec 2025 06:55:19 GMT
 UNIX 1765868119

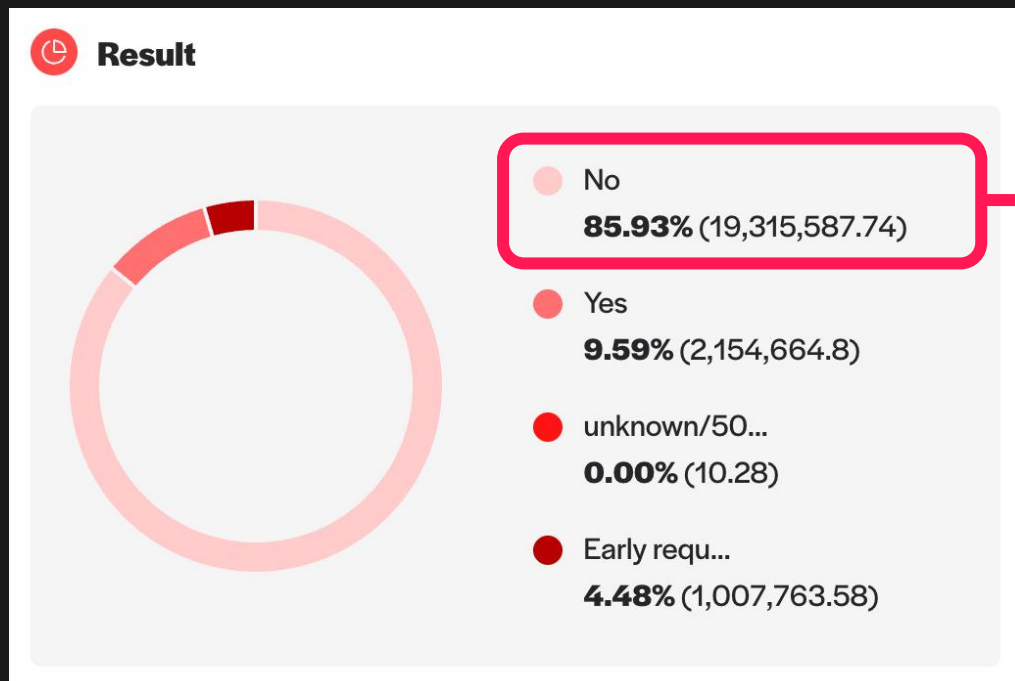
Proposal: No

Bond ⓘ USDC.e 750
 Reward ⓘ USDC.e 5
 Challenge period ends ⓘ 12/16/2025, 5:55 PM

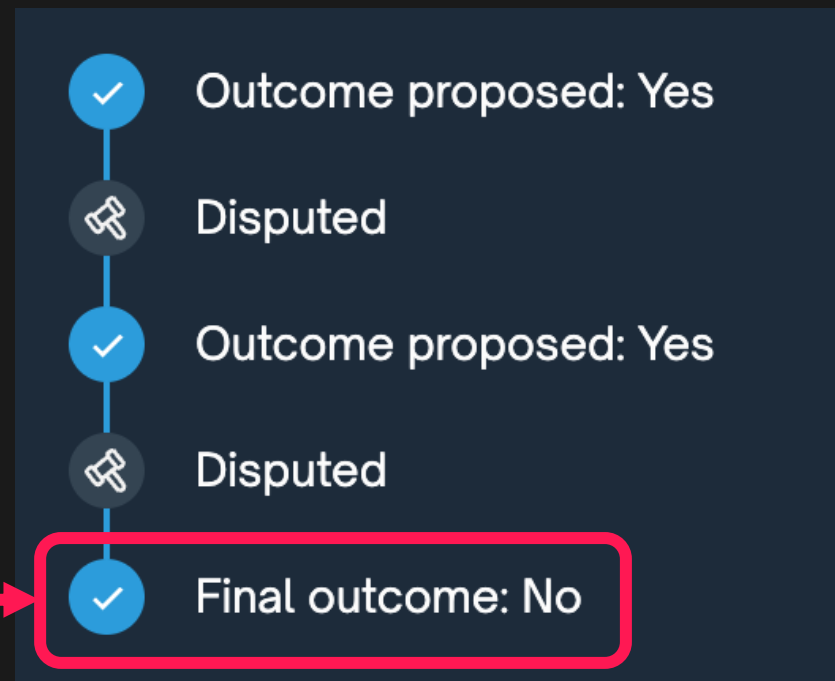
보증금 필요

Oracle & Dispute

마켓이 종료되고 결과에 대한 데이터를 가져오는 단계



거버넌스 투표 결과



예측시장 결론

Threat Modeling

2.1 State Inconsistency in Hybrid CLOB

State Inconsistency in Hybrid CLOB

온체인과 오프체인 간의 상태 불일치로 인한 실행 불확실성

Order Book		Graph	Resolution	+ Rewards	
TRADE YES			PRICE	SHARES	TOTAL
			91¢	21,473.87	\$42,789.76
			90¢	23,902.01	\$23,248.54
			89¢	1,793.09	\$1,736.73
		Asks	88¢	160.09	\$140.88
Last: 88¢		Spread: 1¢			
		Bids	87¢	1,607.37	\$1,398.41
			86¢	2,653.26	\$3,680.21
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			84¢	1,610.00	\$6,261.34

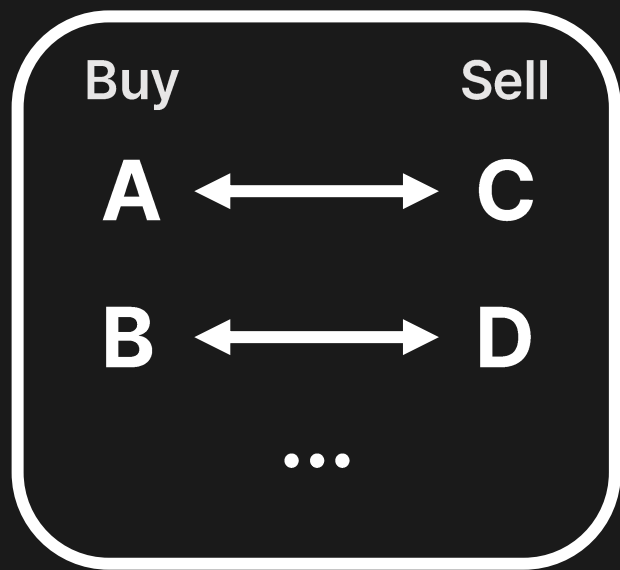
Central Limit Order Book(CLOB)



General-Purpose Chain

State Inconsistency in **Hybrid CLOB**

온체인과 오프체인 간의 상태 불일치로 인한 실행 불확실성



Off-chain : Order Matching

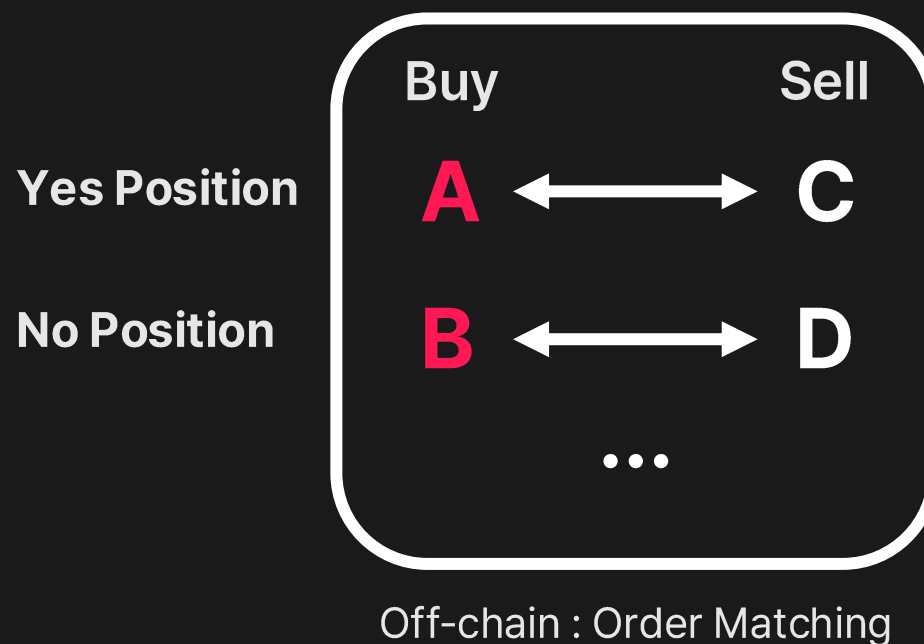


On-chain : Balance Settlement

State Inconsistency in **Hybrid CLOB**

온체인과 오프체인 간의 상태 불일치로 인한 실행 불확실성

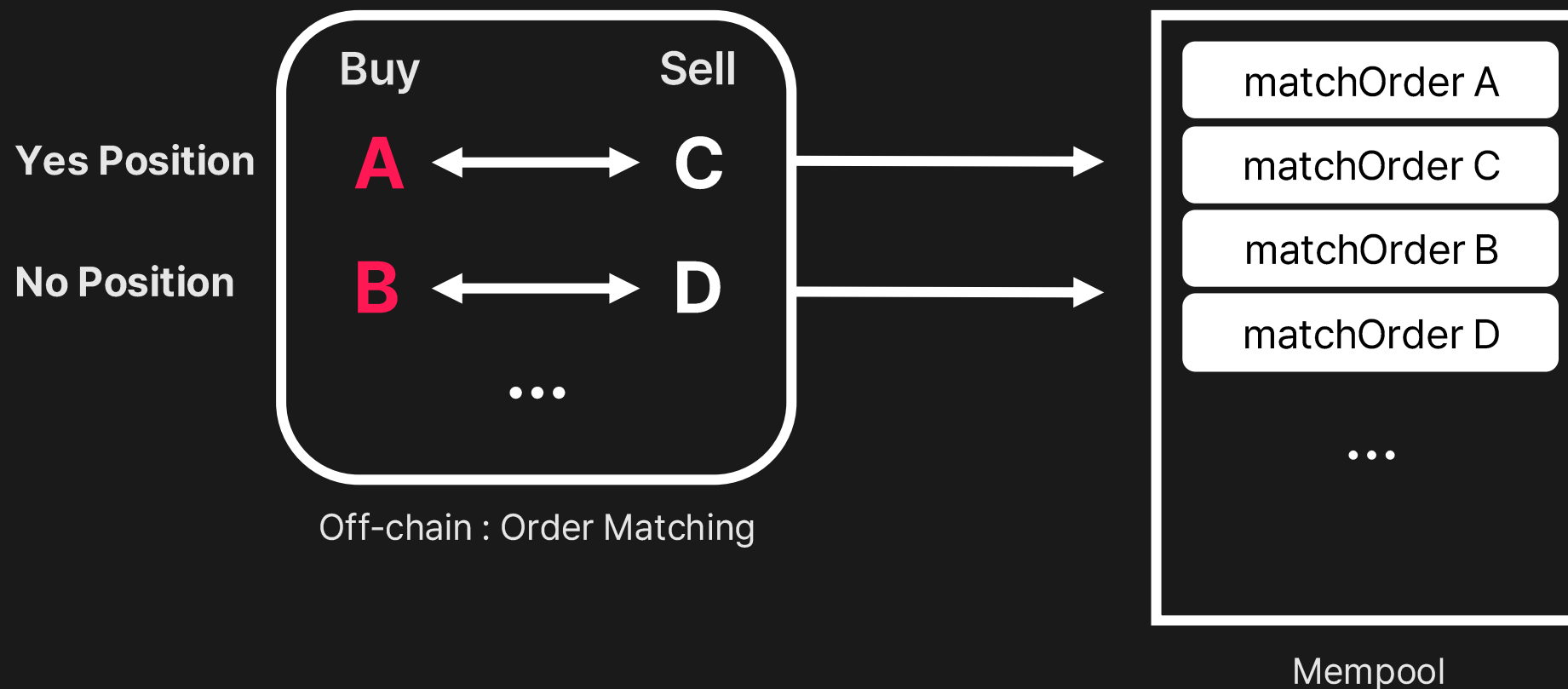
공격 시도 마켓 : Crypto 5m



State Inconsistency in **Hybrid CLOB**

온체인과 오프체인 간의 상태 불일치로 인한 실행 불확실성

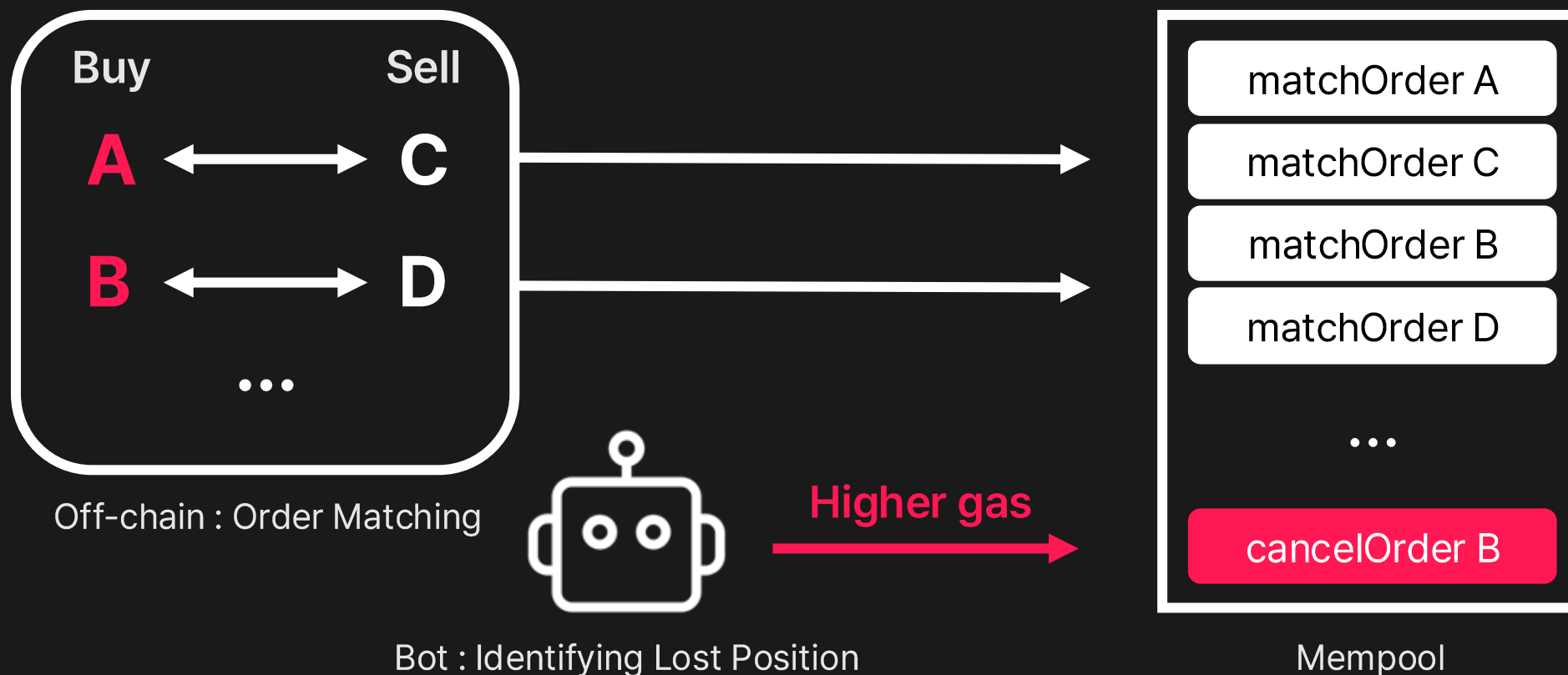
공격 시도 마켓 : Crypto 5m



State Inconsistency in **Hybrid CLOB**

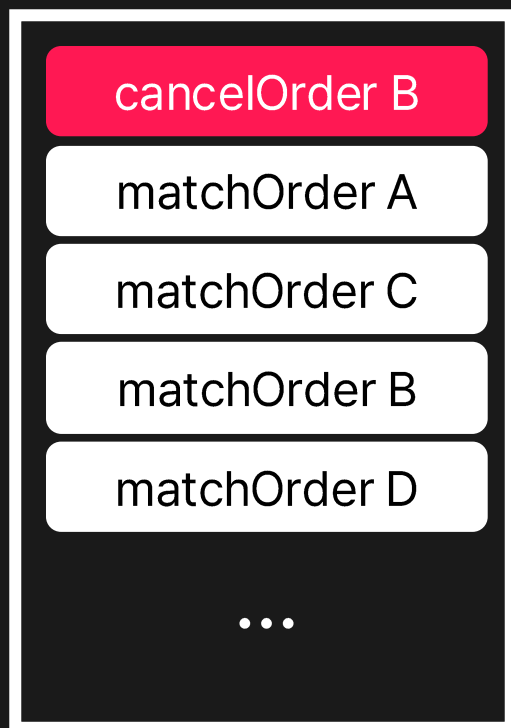
온체인과 오프체인 간의 상태 불일치로 인한 실행 불확실성

공격 시도 마켓 : Crypto 5m



State Inconsistency in **Hybrid CLOB**

온체인과 오프체인 간의 상태 불일치로 인한 실행 불확실성

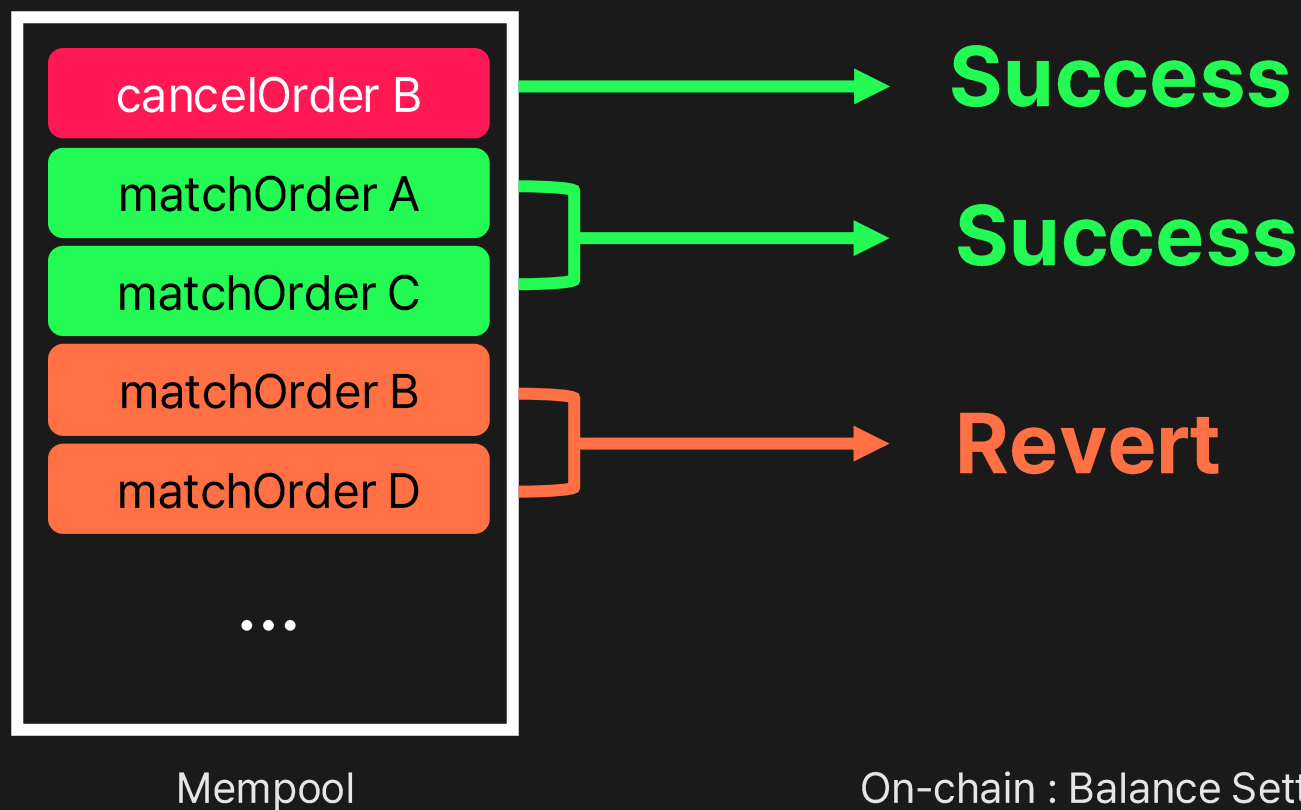


Mempool

On-chain : Balance Settlement

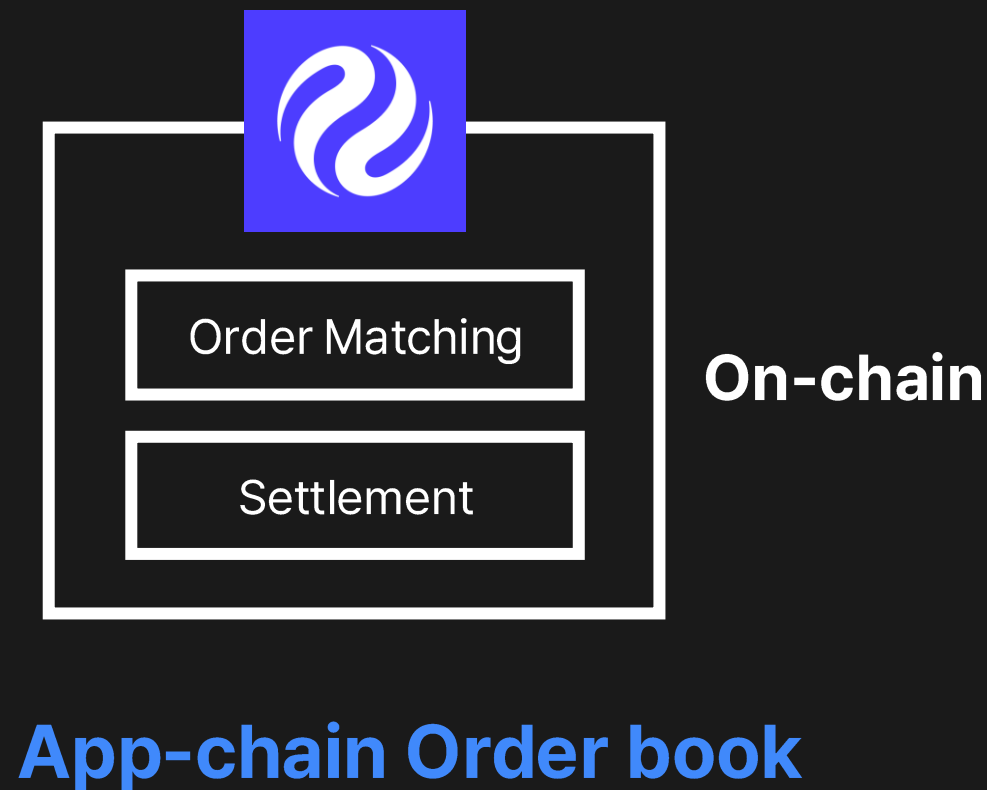
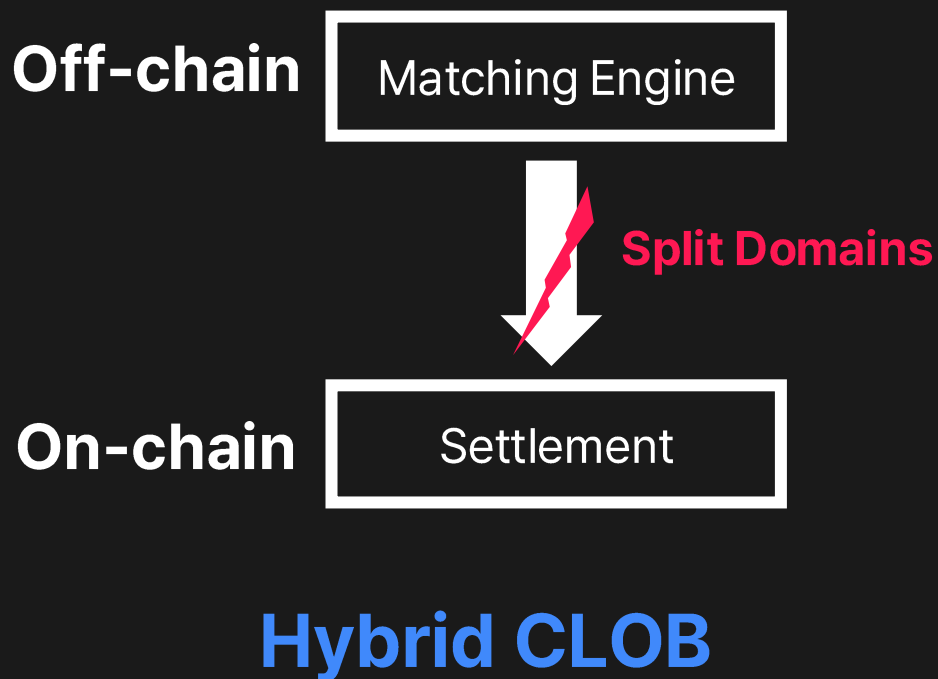
State Inconsistency in **Hybrid CLOB**

온체인과 오프체인 간의 상태 불일치로 인한 실행 불확실성



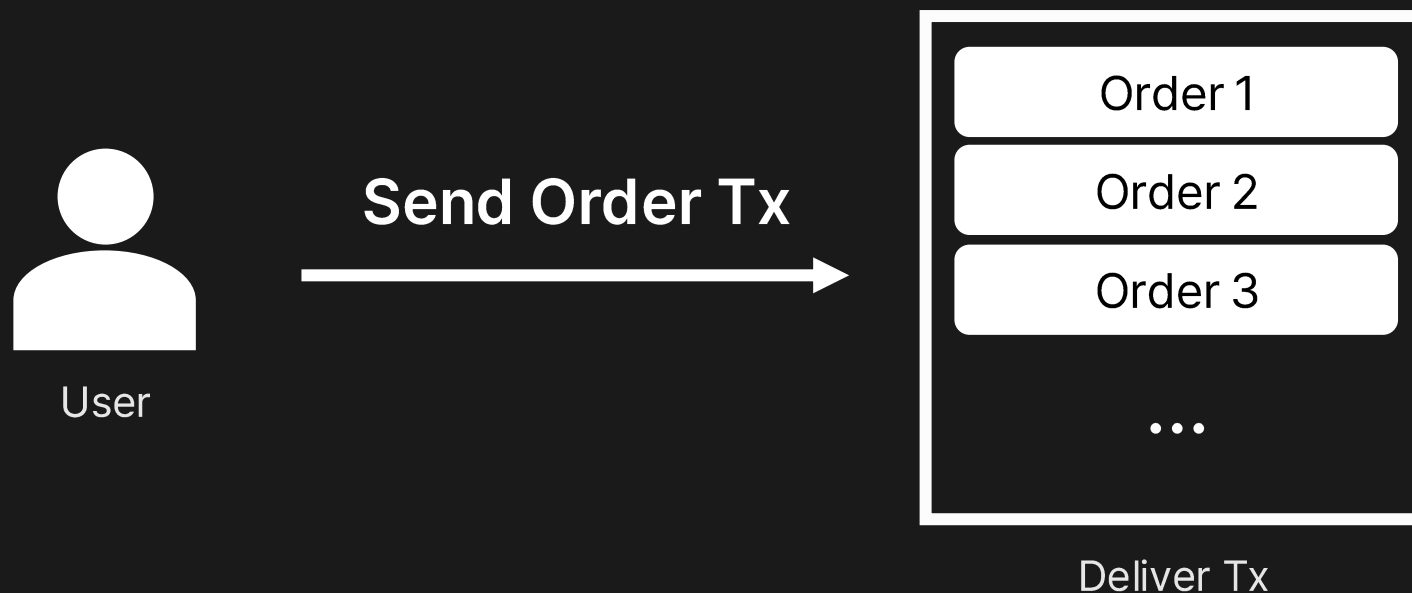
Mitigation : App-chain based On-chain Order book

온체인에 오더북을 올려 Hybrid CLOB 문제점 해결



Mitigation : App-chain based On-chain Order book

온체인에 오더북을 올려 Hybrid CLOB 문제점 해결



Mitigation : App-chain based On-chain Order book

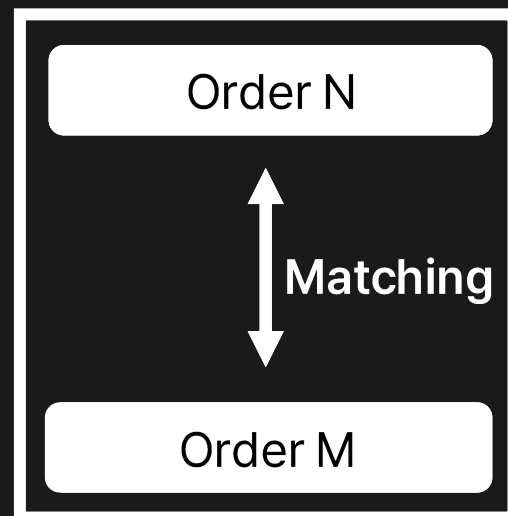
온체인에 오더북을 올려 Hybrid CLOB 문제점 해결



FBA: Frequent Batch Auction

Mitigation : App-chain based On-chain Order book

온체인에 오더북을 올려 Hybrid CLOB 문제점 해결



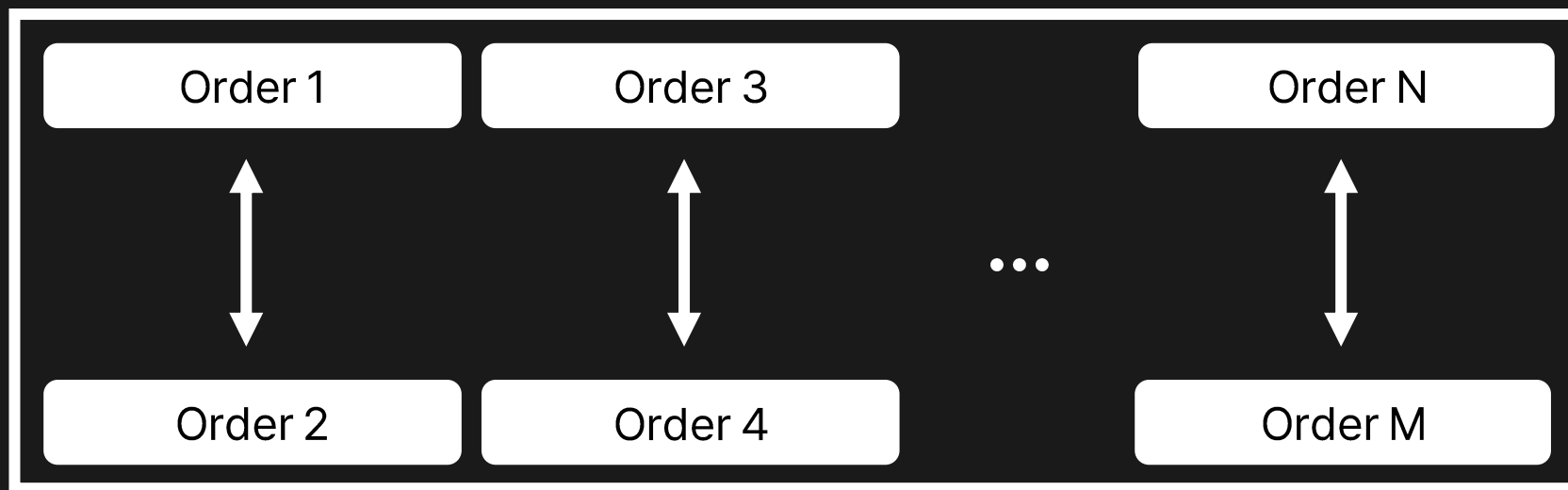
EndBlock: FBA Matching

Settlement
Position & Balance

FBA: Frequent Batch Auction

Mitigation : App-chain based On-chain Order book

온체인에 오더북을 올려 Hybrid CLOB 문제점 해결

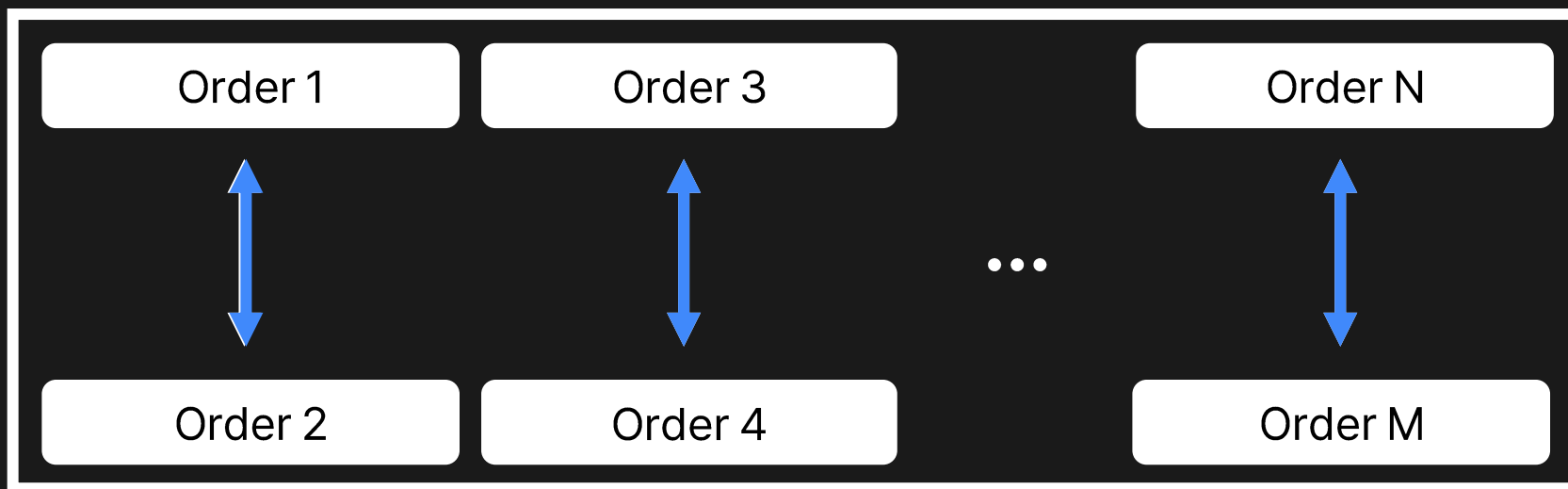


EndBlock: FBA Matching

FBA: Frequent Batch Auction
UCP: Uniform Clearing Price

Mitigation : App-chain based On-chain Order book

온체인에 오더북을 올려 Hybrid CLOB 문제점 해결

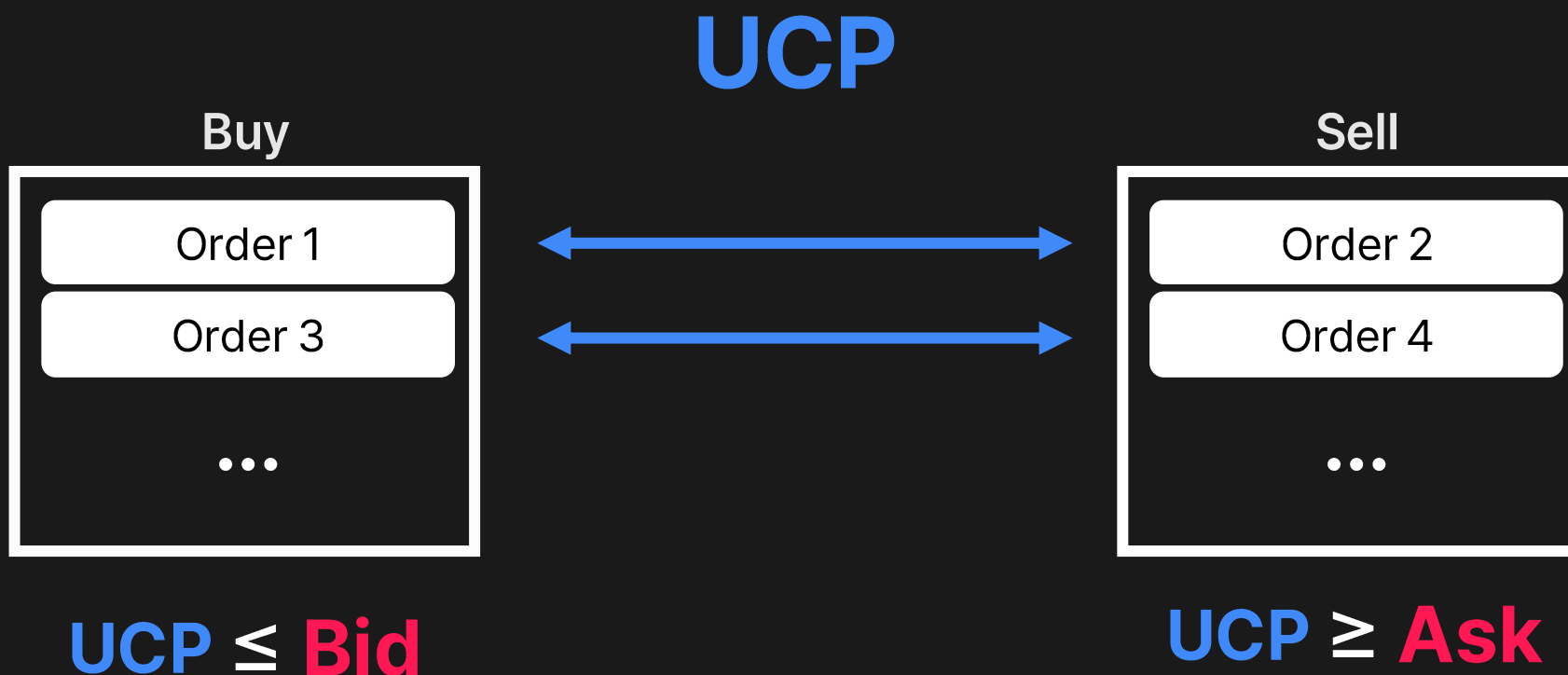


EndBlock: FBA Matching

Equal Price
Matching

Mitigation : App-chain based On-chain Order book

온체인에 오더북을 올려 Hybrid CLOB 문제점 해결



FBA: Frequent Batch Auction
UCP: Uniform Clearing Price

Threat Modeling

2.2 Single External Data Source Failure

Single External Data Source Failure

단일 데이터 소스를 유일한 판정 기준으로 삼아 룰북 설계 문제 발생

Rules

This market will resolve to the calendar date (ET) of the first day which the U.S. The Office of Personnel Management (OPM) announces that the U.S. federal government is not shut down due to a lapse in appropriations.

A qualifying announcement date will determine this market's resolution, even if OPM lists a later reopening date (e.g., if OPM announces on November 10 that the government will reopen on November 11, the market will resolve to November 10).

Partial shutdowns count as shutdowns; announcements of office closures due to holidays or inclement weather do not qualify as a shutdown.

The resolution source for this market will be OPM's Operating Status page (<https://www.opm.gov/policy-data-oversight/snow-dismissal-procedures/current-status/>).

Created At: Nov 10, 2025, 10:35 AM

Single External Data Source Failure

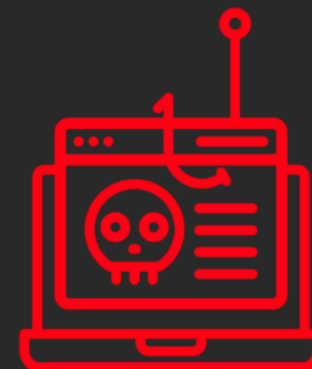
단일 데이터 소스를 유일한 판정 기준으로 삼아 룰북 설계 문제 발생



DDoS



데이터 위변조



DNS 하이재킹

Single External Data Source Failure

단일 데이터 소스를 유일한 판정 기준으로 삼아 룰북 설계 문제 발생

Rules

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Partial shutdowns count as shutdowns; announcements of office closures due to holidays or inclement weather do not qualify as a shutdown.

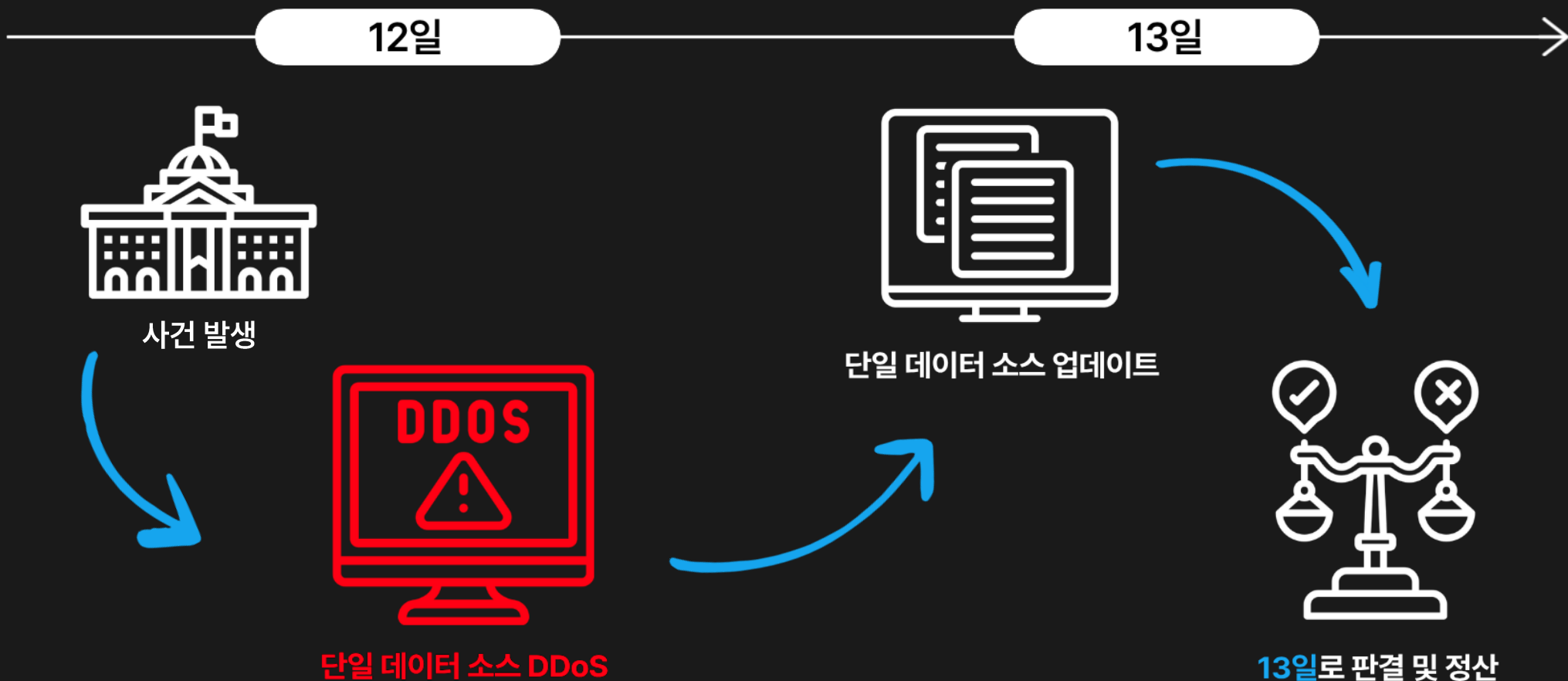
The resolution source for this market will be OPM's Operating Status page (<https://www.opm.gov/policy-data-oversight/snow-dismissal-procedures/current-status/>).

Created At: Nov 10, 2025, 10:35 AM

실제 사건 발생일과 무관하게 웹사이트 글 업데이트 날짜로 결정되게 룰북 설정

Single External Data Source Failure

단일 데이터 소스를 유일한 판정 기준으로 삼아 룰북 설계 문제 발생



Mitigation : Multi-Data Source

콜백에 다중 데이터 소스 링크 설정



One Market

SPOF



Single Source

Mitigation : Multi-Data Source

롤북에 다중 데이터 소스 링크 설정



One Market



Multi Source

Mitigation : Multi-Data Source

룰북에 다중 데이터 소스 링크 설정



Official Source



Credible Media

Mitigation : Multi-Data Source

룰북에 다중 데이터 소스 링크 설정



P of Q

Mitigation : Multi-Data Source

단일 Web2 데이터 소스에 대한 의존 제거



DDoS



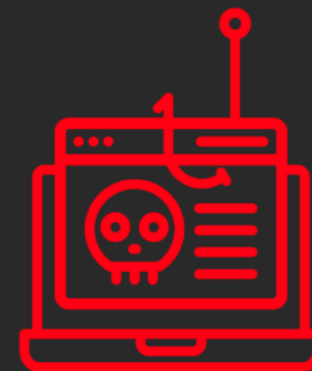
공격 비용 증가



데이터 위변조



해킹 난이도 상승



DNS 하이재킹



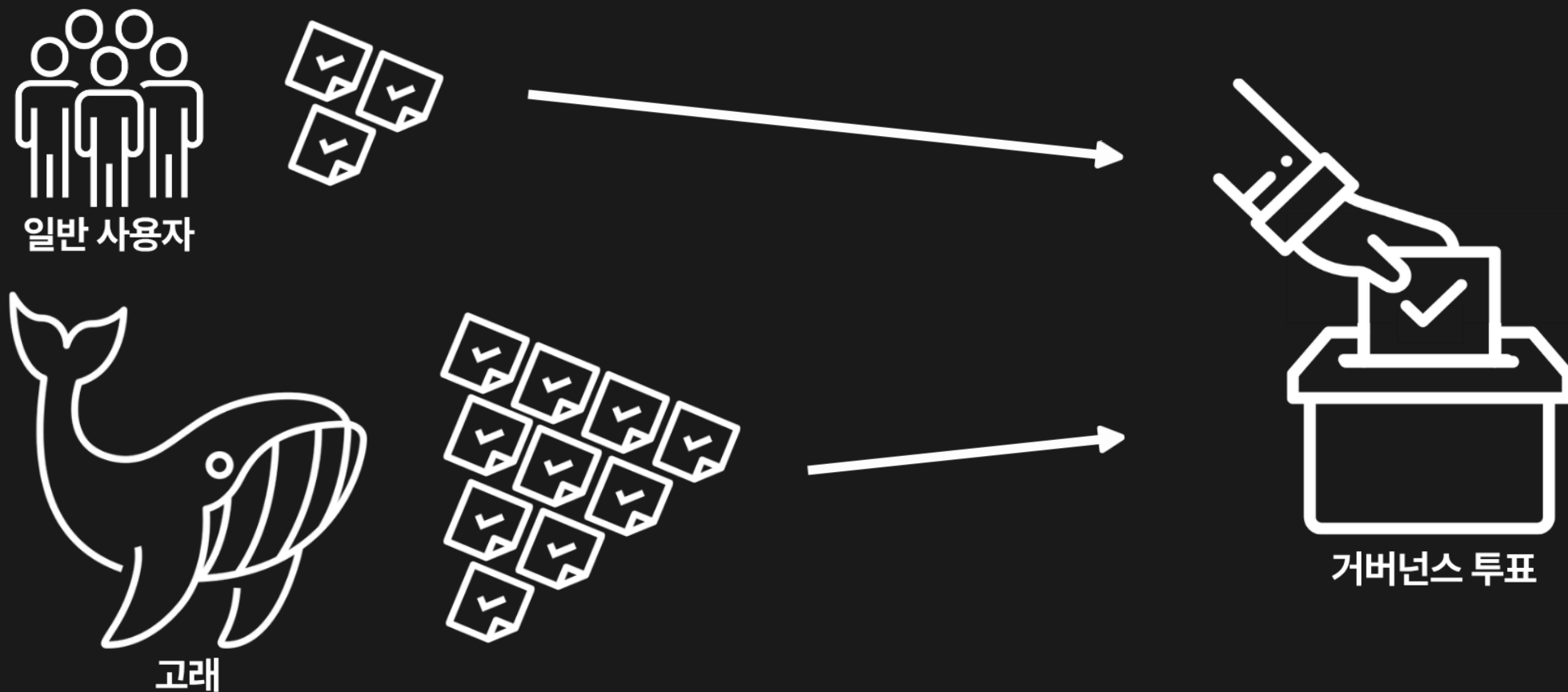
경로 다양성

Threat Modeling

2.3 Whale Attack 51%

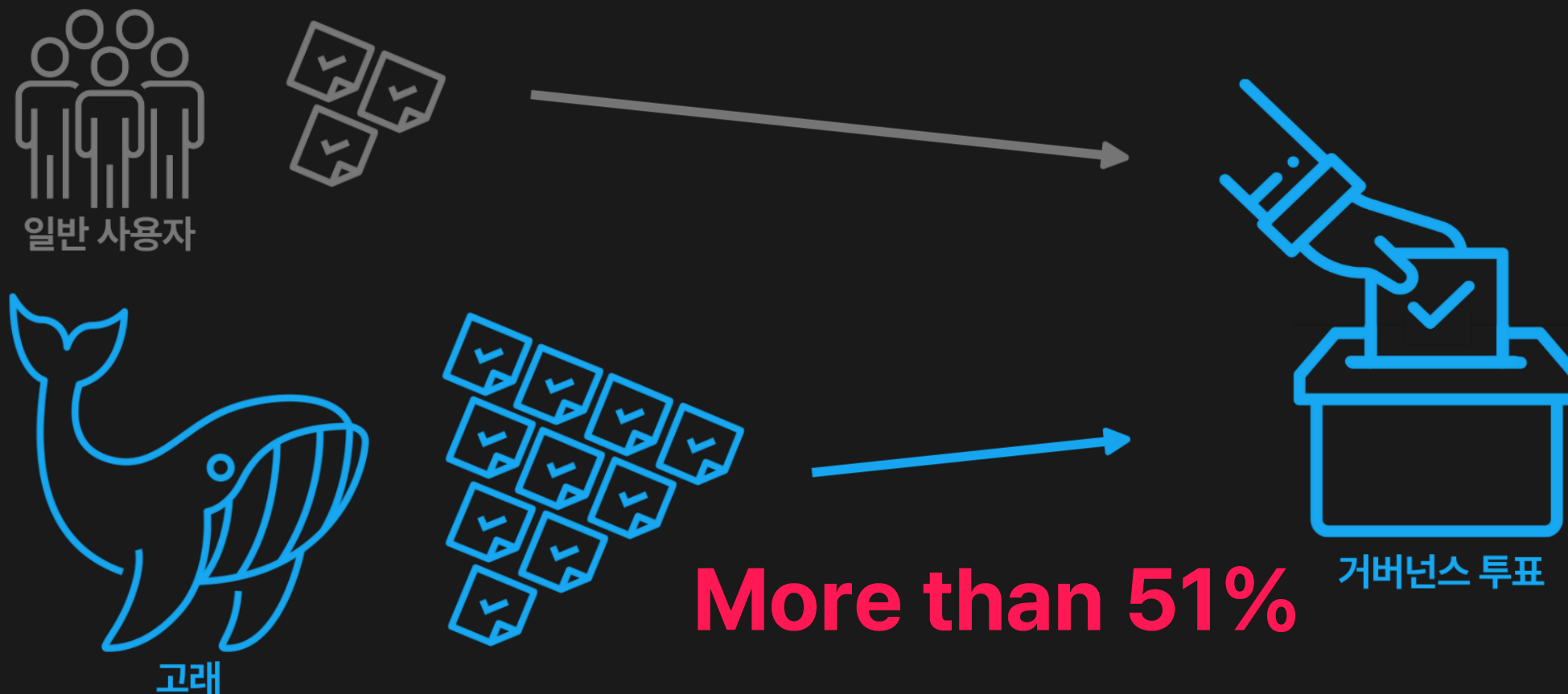
Whale Attack 51%

고래가 투표 결과를 통제함으로써 시장 결론을 왜곡할 수 있는 위험



Whale Attack 51%

고래가 투표 결과를 통제함으로써 시장 결론을 왜곡할 수 있는 위험



Whale Attack 51% – Analysis

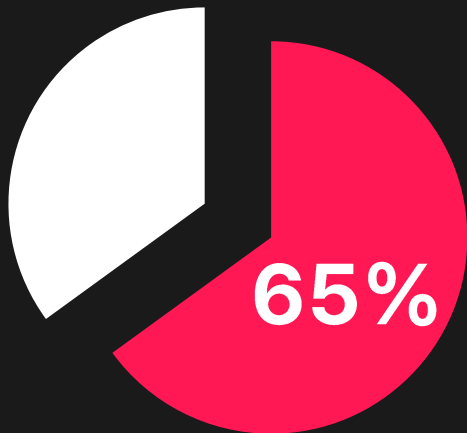
UMA 오라클 조작 공격의 경제적 성립 조건 (Profit > Cost)

시장 결과가 나고 공격자가 얻는 최종적 이익 계산

공격자 순이익 = 공격으로 얻는 이익(Profit) - 51% 공격비용(Cost)

Whale Attack 51% – Analysis

공격 Cost의 고정 및 하락: '단일 라운드 참여량' 기준



총 스테이킹 UMA의 65% 이상



```
struct Round {
    SlashingLibraryInterface slashingLibrary; // Slashing l
    uint128 minParticipationRequirement; // Minimum staked
    uint128 minAgreementRequirement; // Minimum staked toke
    uint128 cumulativeStakeAtRound; // Total staked tokens
    uint32 numberOfRequestsToVote; // The number of request
}
```

3. cumulativeStake (0xdd487283)( Public Variable)

17579675449120743245465290 uint128

약 1,758만 UMA

65%

약 1,143만 UMA

Whale Attack 51% – Analysis

공격 Cost의 고정 및 하락: '단일 라운드 참여량' 기준



약 1,143만 × \$0.4 = 약 457만 달러

Whale Attack 51% – Analysis

공격 Profit의 확장성: 다중 시장 Batching 악용

Active votes:

Time remaining to commit votes: 18 hours

Reveal phase starts in: 18 hours

Remind me

Vote	Your vote	Vote status
 Will another app be #1 Free App in Play Store on December 19? Polymarket 2025-12-20, 2:00 AM	No	Not committed
 Will ChatGPT be #1 Free App in Play Store on December 19? Polymarket 2025-12-20, 2:00 AM	Yes	Not committed
 Ethereum Up or Down - December 18, 3PM ET Polymarket 2025-12-19, 6:00 AM	Up	Not committed
 Will the price of Bitcoin be between \$86,000 and \$88,000 on December 18? Polymarket 2025-12-19, 2:04 AM	Yes	Not committed
 Will the price of Bitcoin be less than \$80,000 on December 18? Polymarket 2025-12-19, 2:04 AM	Yes	Not committed
 Will the highest temperature in Dallas be 59°F or below on December 18? Polymarket 2025-12-18, 3:20 PM	Yes	Not committed
 Will the highest temperature in Seoul be 6°C on December 18? Polymarket 2025-12-18, 1:58 PM	Yes	Not committed
 Bryan College Lions vs. Samford Bulldogs Polymarket 2025-12-18, 11:40 AM	Samford Bulldogs	Not committed

단일 투표 라운드 3일간
다수의 시장 동시 처리

Whale Attack 51% – Analysis

공격 Profit의 확장성: 다중 시장 Batching 악용

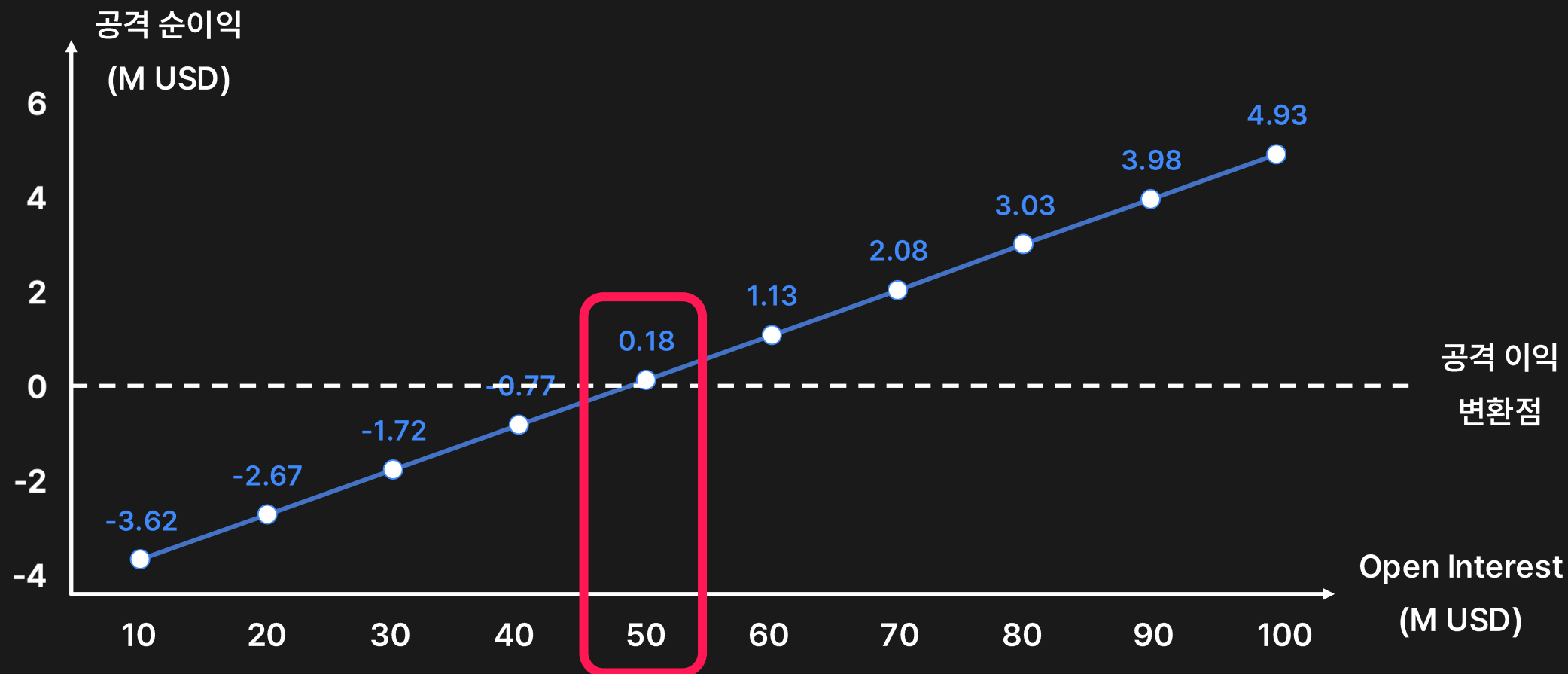
Market	Source	Volume	Open Interest	Ends	Tags
+ Netanyahu out by...?	Polymarket	\$86,230,103	\$30,703,771	12/31/2026	Earn 4% +5
+ Presidential Election Winner 2028	Polymarket	\$16,224,038	\$28,592,049	11/07/2028	Earn 4% +7
+ Will the Iranian regime fall by March 31?	Polymarket	\$61,993,151	\$16,821,672	03/31/2026	Geopolitics +8
+ US x Iran ceasefire by...?	Polymarket	\$55,945,699	\$14,123,303	01/01/1970	Geopolitics +5
+ US forces enter Iran by...?	Polymarket	\$57,908,858	\$13,510,929	01/01/1970	Geopolitics +3
+ Democratic Presidential Nominee 2028	Polymarket	\$17,054,345	\$8,004,456	11/07/2028	Earn 4% +7
+ Will the Iranian regime fall by June 30?	Polymarket	\$23,301,707	\$7,831,814	06/30/2026	Geopolitics +8
+ Republican Presidential Nominee 2028	Polymarket	\$14,253,330	\$7,602,450	11/07/2028	Earn 4% +7
+ Will the Iranian regime fall by April 30?	Polymarket	\$13,429,930	\$6,207,983	04/30/2026	Geopolitics +6
+ Will Trump visit China by...?	Polymarket	\$10,337,052	\$5,896,888	04/30/2026	China +8
+ Trump out as President by March 31?	Polymarket	\$13,429,840	\$5,883,882	03/31/2026	Elections +3
+ What price will Bitcoin hit in March?	Polymarket	\$39,628,298	\$5,488,322	04/01/2026	Bitcoin +5
+ Will the US confirm that aliens exist by...?	Polymarket	\$20,620,083	\$5,220,879	12/31/2026	Aliens +3
+ Russia x Ukraine ceasefire by end of 2026?	Polymarket	\$12,439,734	\$4,681,737	12/31/2026	Earn 4% +7

Attack Scenario:

Worst-Case로 이 상위 시장들을
동시다발적 dispute로 batching 조작

Whale Attack 51% – Analysis

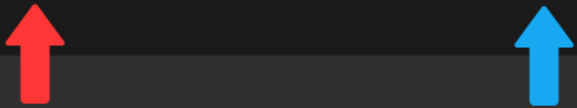
공격 이익과 공격 비용의 상관관계를 정량적으로 계산



Whale Attack 51% – Analysis

UMA 오라클 조작 공격의 경제적 성립 조건 (Profit > Cost)

시장 결과가 나고 공격자가 얻는 최종적 이익 계산


$$\text{공격자 순이익} = \text{공격으로 얻는 이익(Profit)} - 51\% \text{ 공격비용(Cost)}$$

Mitigation : Dynamic Dual-Staking to Raise Attack Cost

투표 라운드에서 총 이가 커질수록 ETH 스테이킹 요구치를 자동으로 높여 비율 조정



UMA 토큰 1개

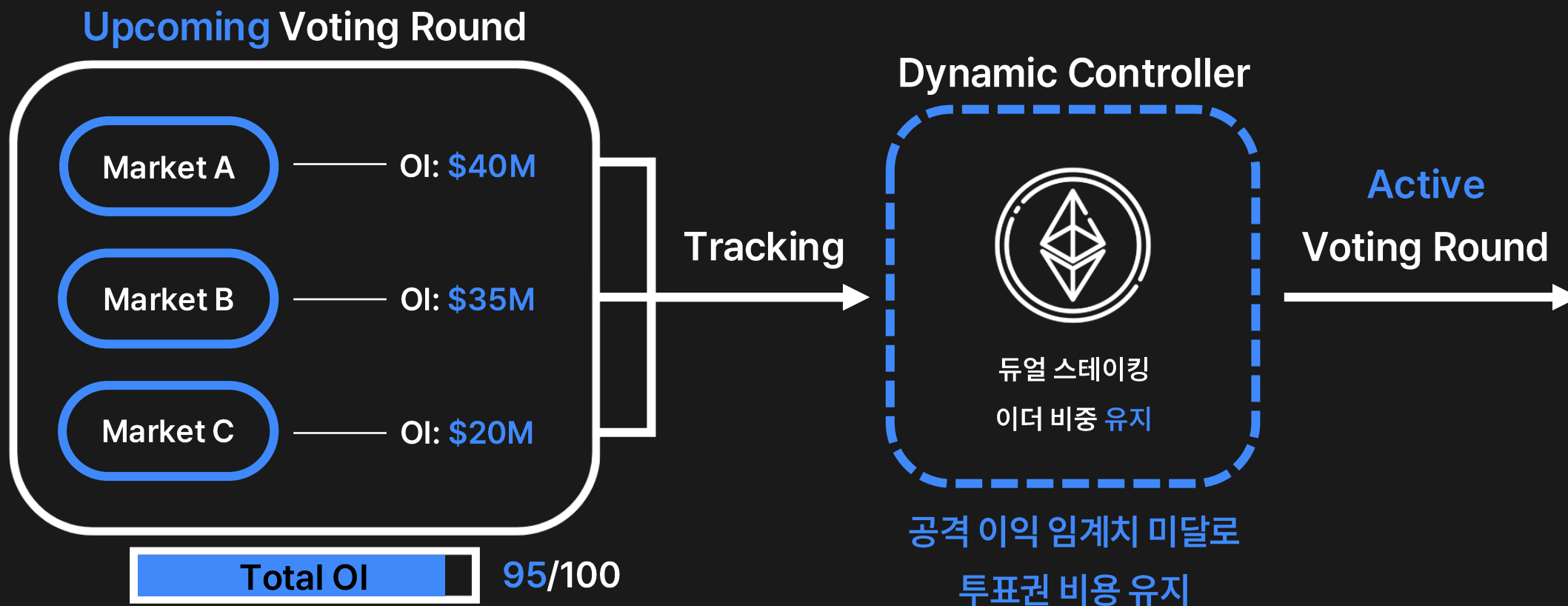
ETH 토큰 비율 조정
ex) 0.0002 ether

투표권 1개

공격 비용 2배 증가

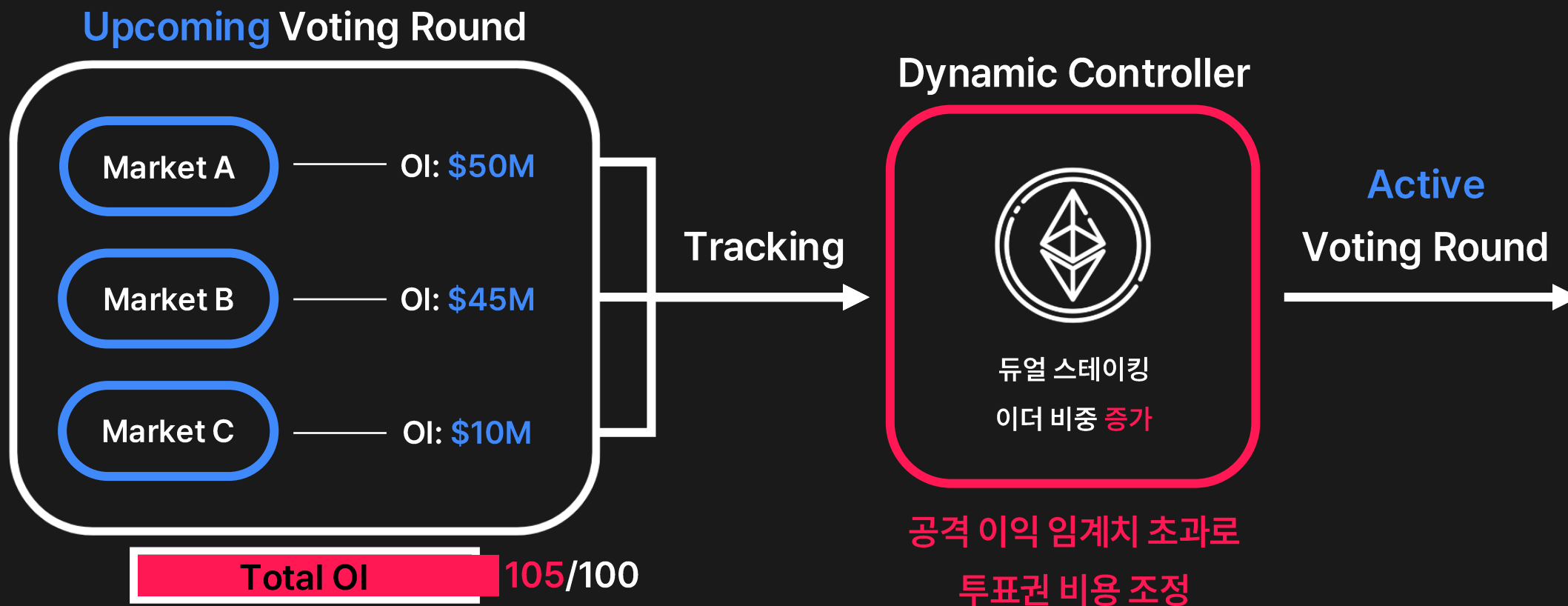
Mitigation : Dynamic Dual-Staking to Raise Attack Cost

투표 라운드에서 총 이가 커질수록 ETH 스테이킹 요구치를 자동으로 높여 비율 조정



Mitigation : Dynamic Dual-Staking to Raise Attack Cost

투표 라운드에서 총 이가 커질수록 ETH 스테이킹 요구치를 자동으로 높여 비율 조정

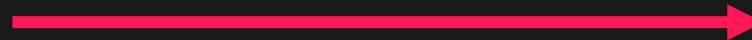


Mitigation : Dynamic Dual-Staking to Raise Attack Cost

투표 라운드에서 총 이가 커질수록 ETH 스테이킹 요구치를 자동으로 높여 비율 조정



Tracking 공개



투명한 비용 조정



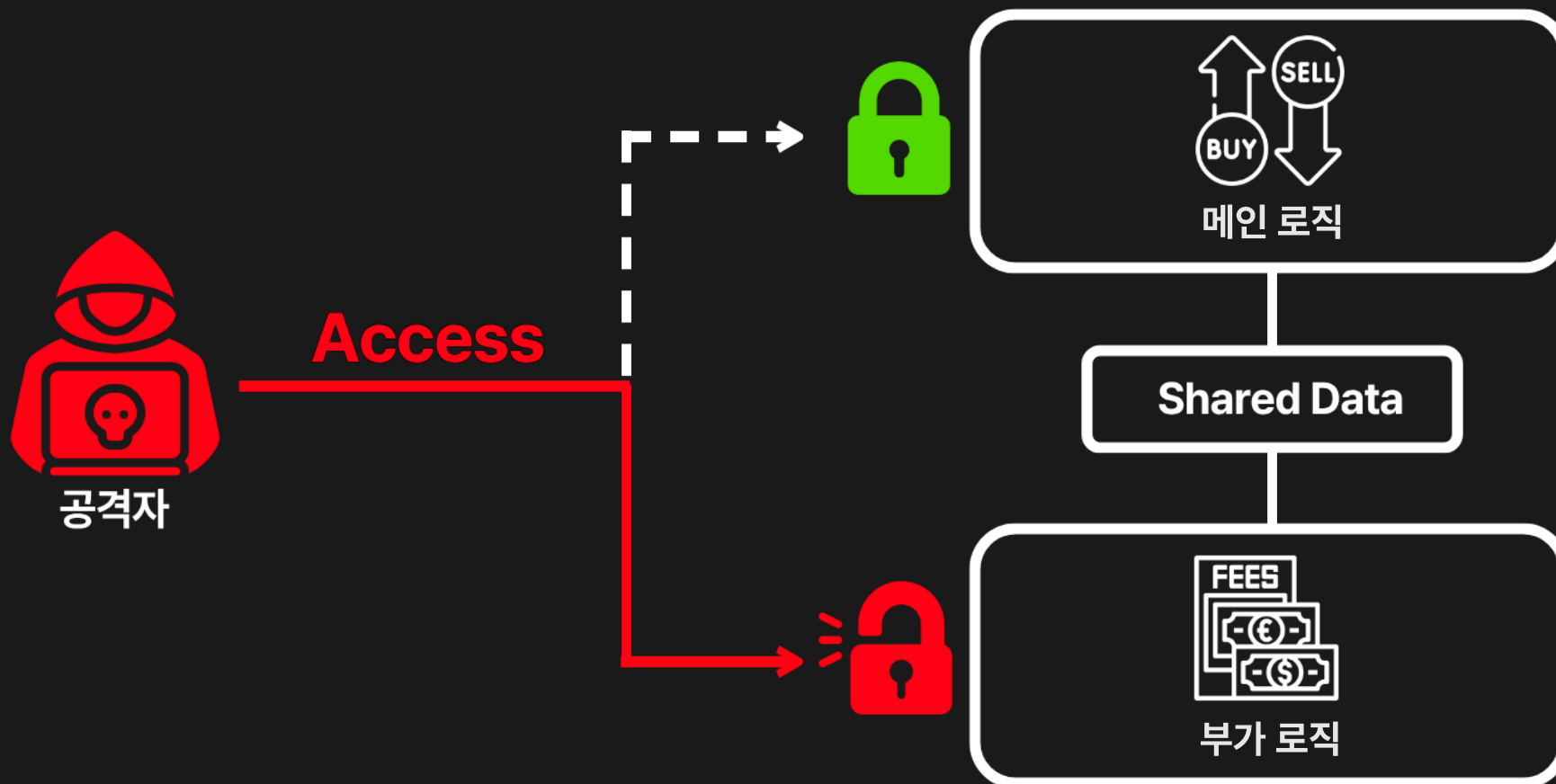
사용자 신뢰성 향상

Threat Modeling

2.4 DoS via Unrestricted Shared State

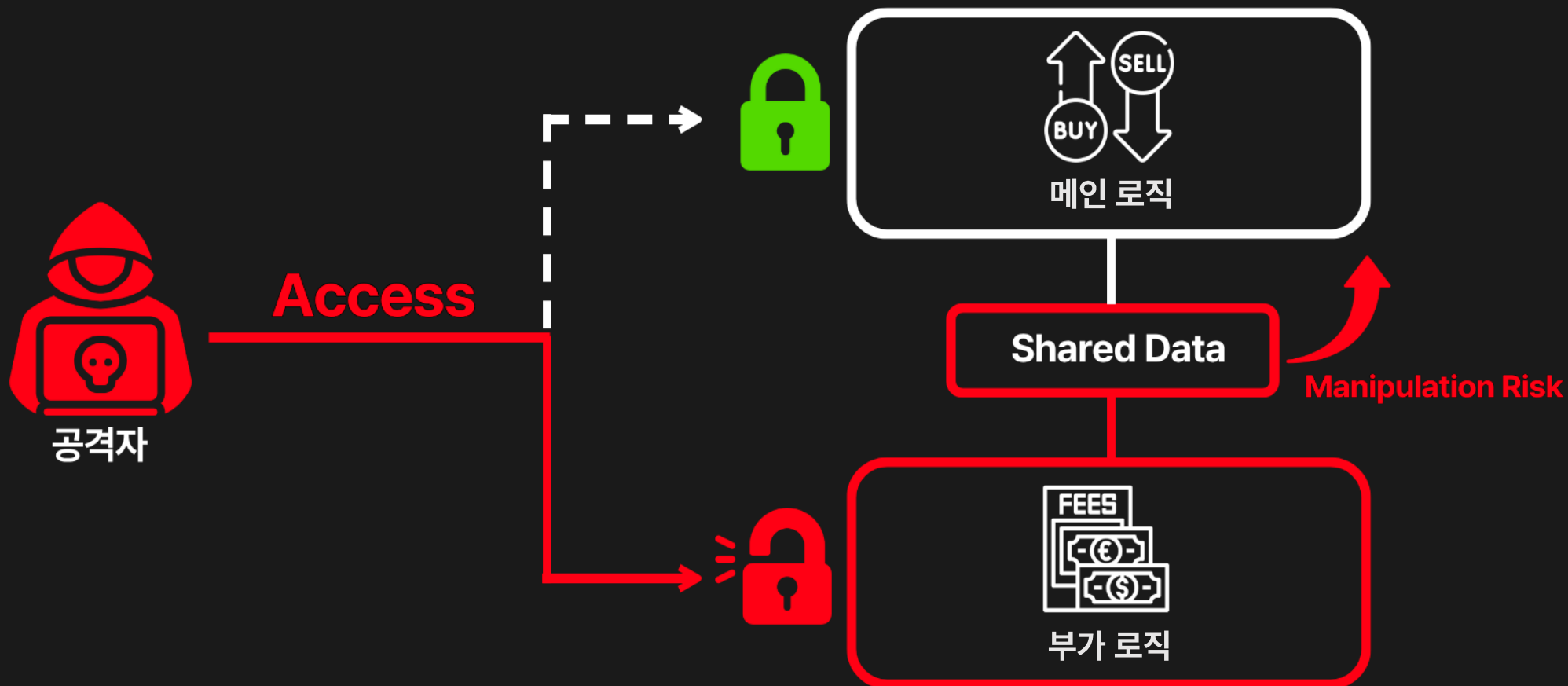
DoS via **Unrestricted Shared State** Manipulation

메인 로직에서 사용하는 데이터가 조작되어 전체 흐름 중단



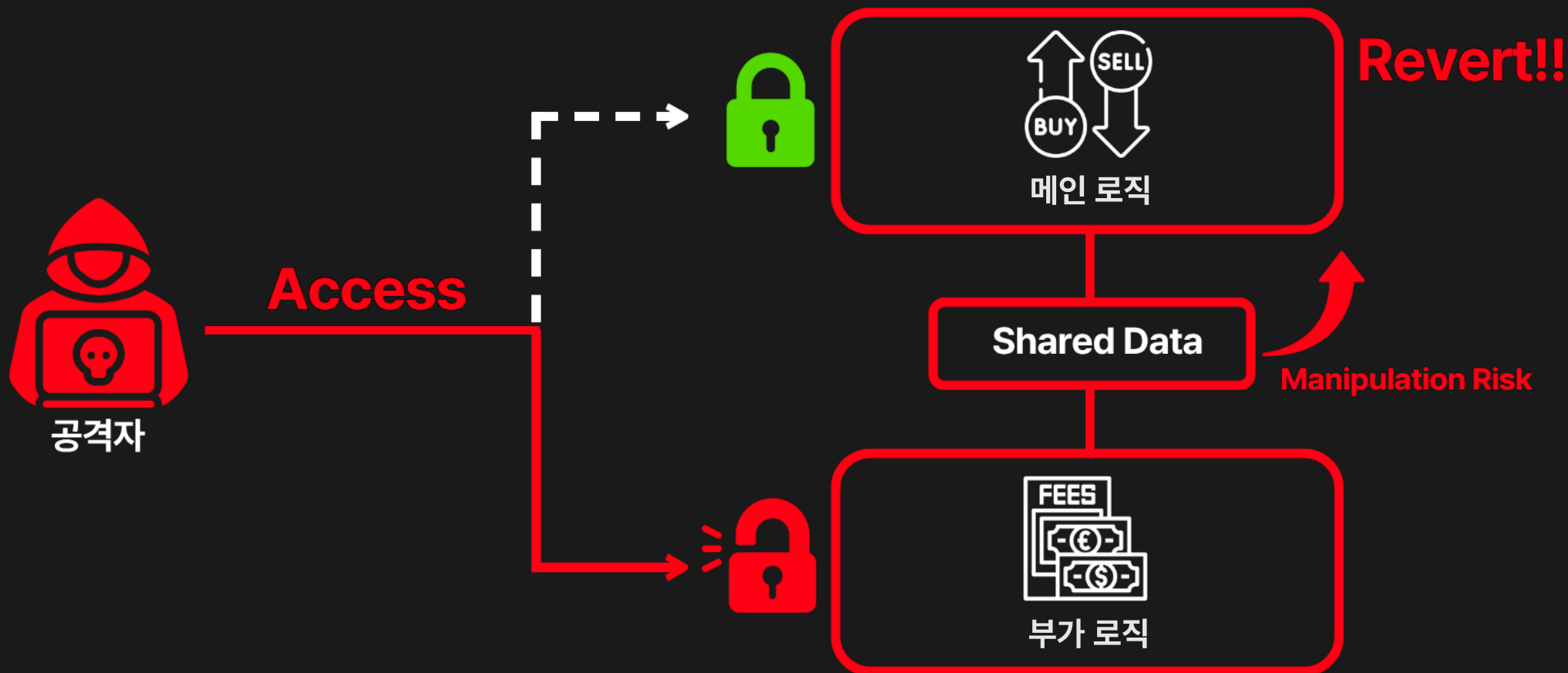
DoS via **Unrestricted Shared State** Manipulation

메인 로직에서 사용하는 데이터가 조작되어 전체 흐름 중단



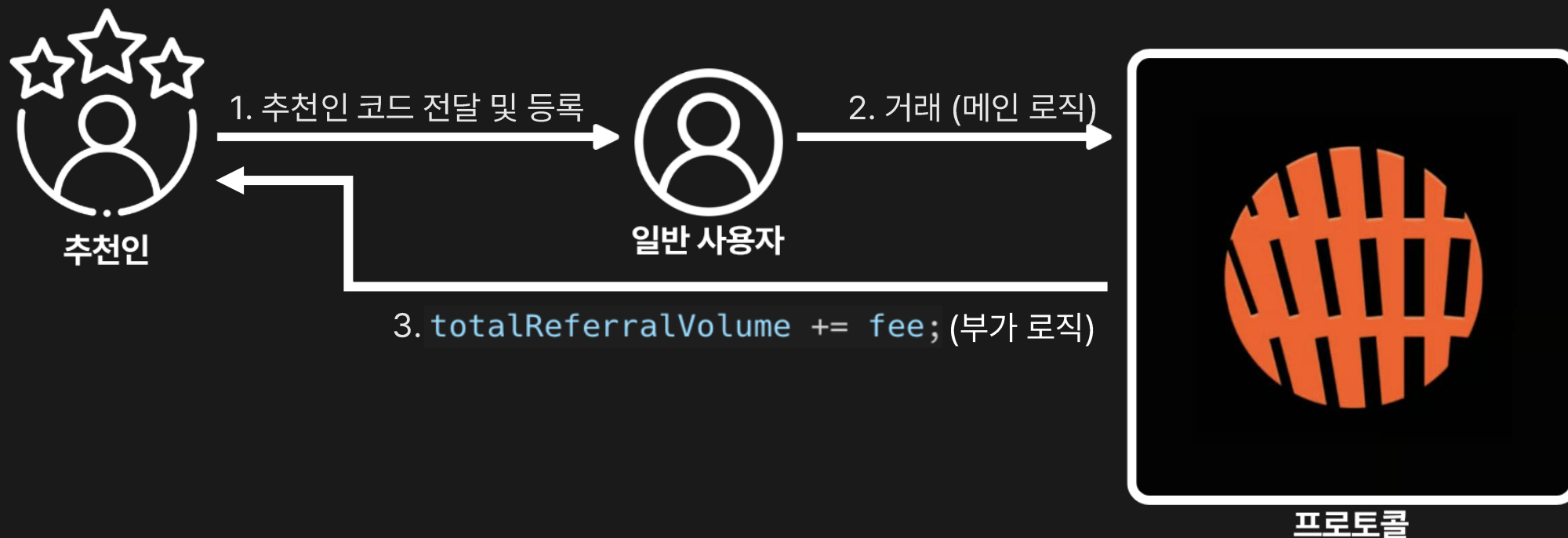
DoS via **Unrestricted Shared State** Manipulation

메인 로직에서 사용하는 데이터가 조작되어 전체 흐름 중단



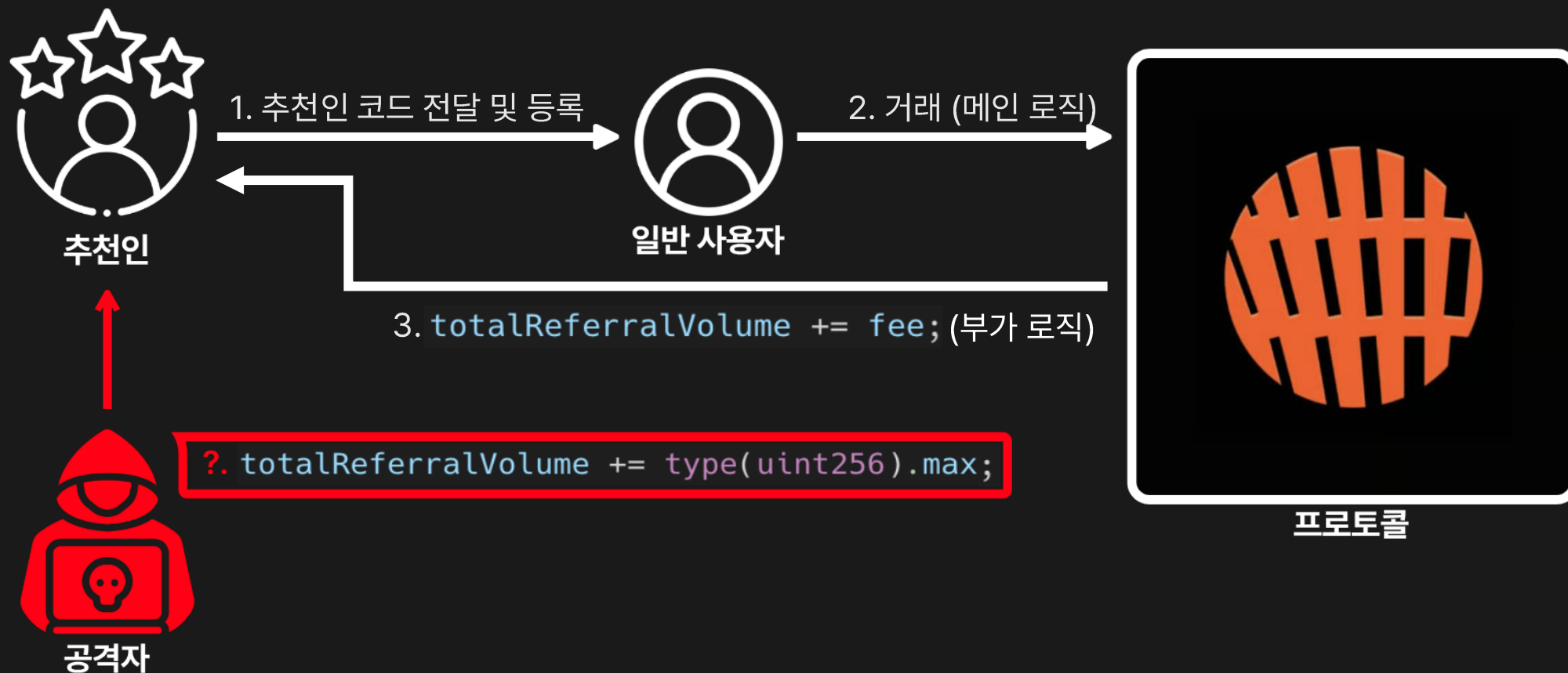
DoS via **Unrestricted Shared State** Manipulation

거래 로직이 공유 데이터에 의존하는 경우, 해당 데이터가 조작되면 전체 거래 중단 위험



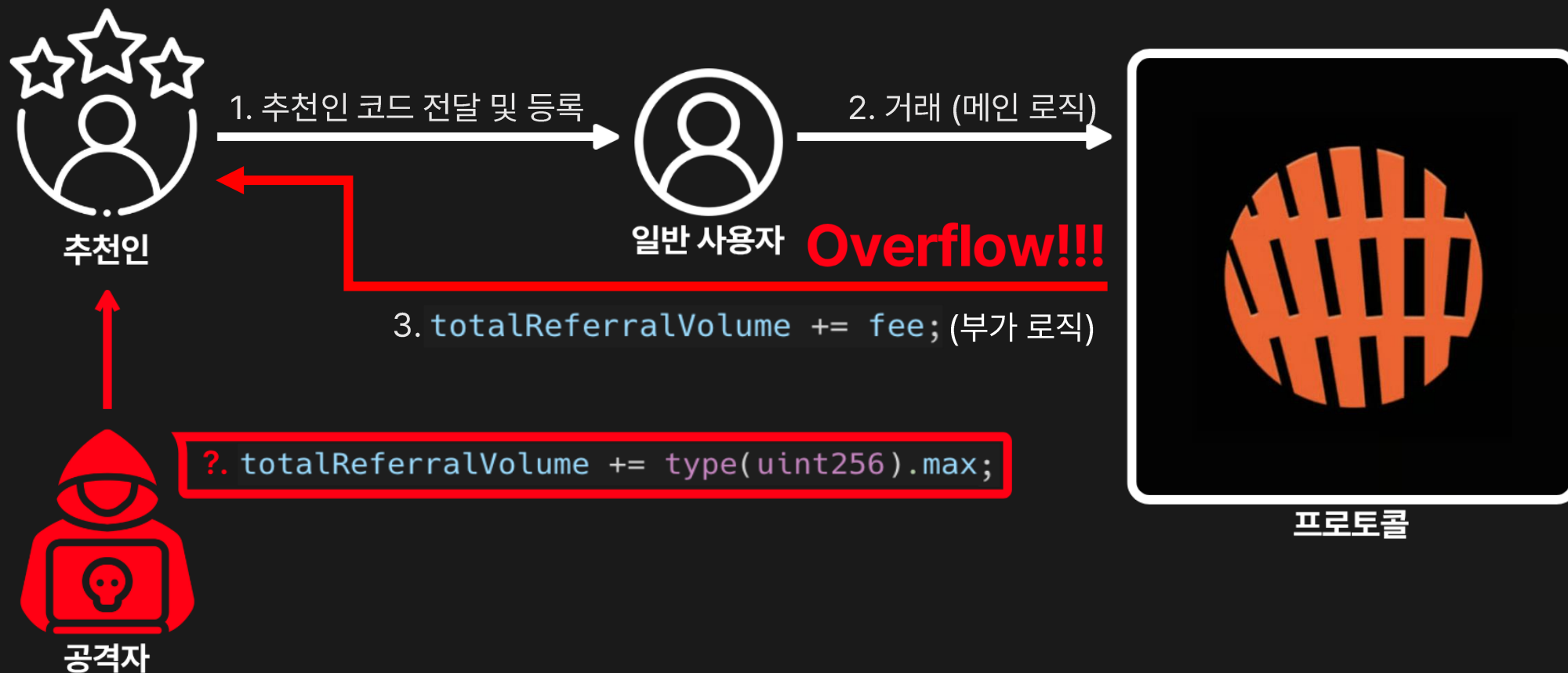
DoS via **Unrestricted Shared State** Manipulation

거래 로직이 공유 데이터에 의존하는 경우, 해당 데이터가 조작되면 전체 거래 중단 위험



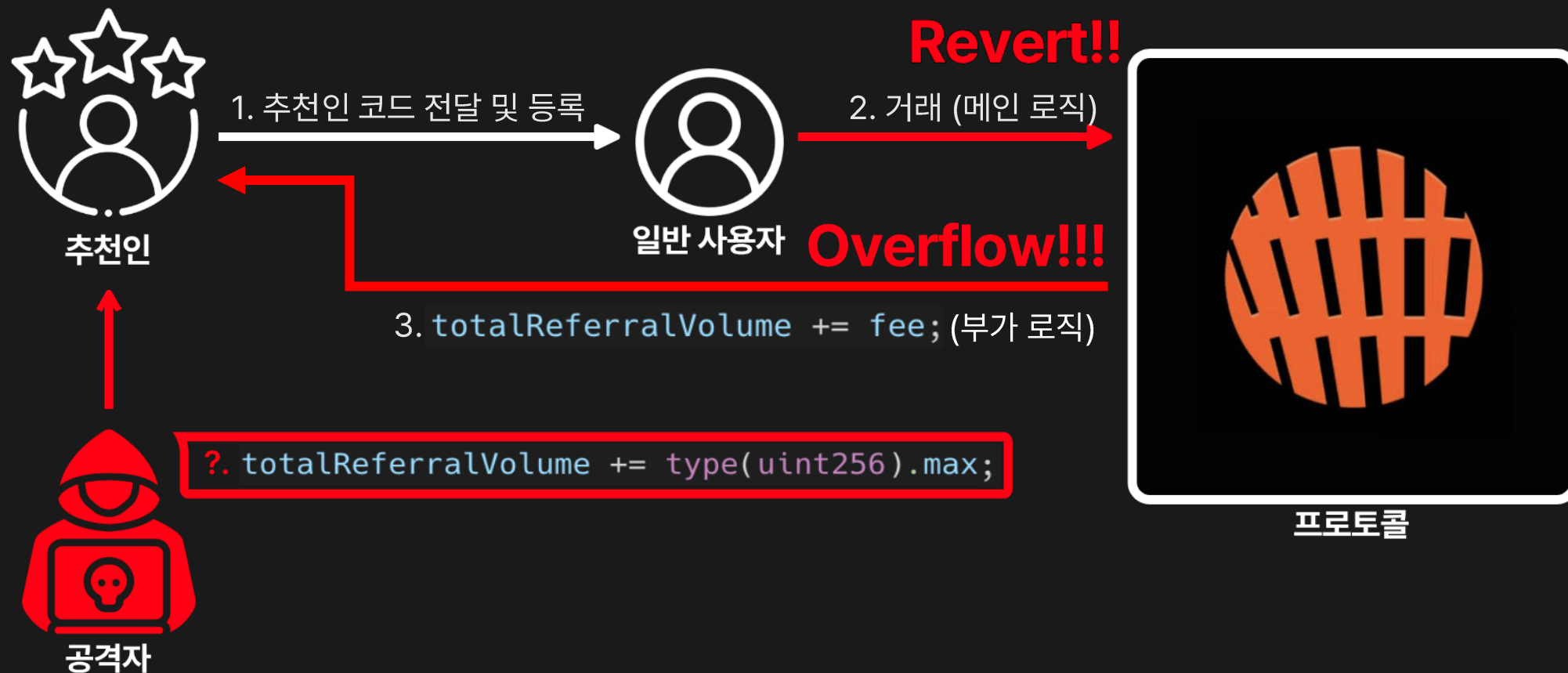
DoS via **Unrestricted Shared State** Manipulation

거래 로직이 공유 데이터에 의존하는 경우, 해당 데이터가 조작되면 전체 거래 중단 위험



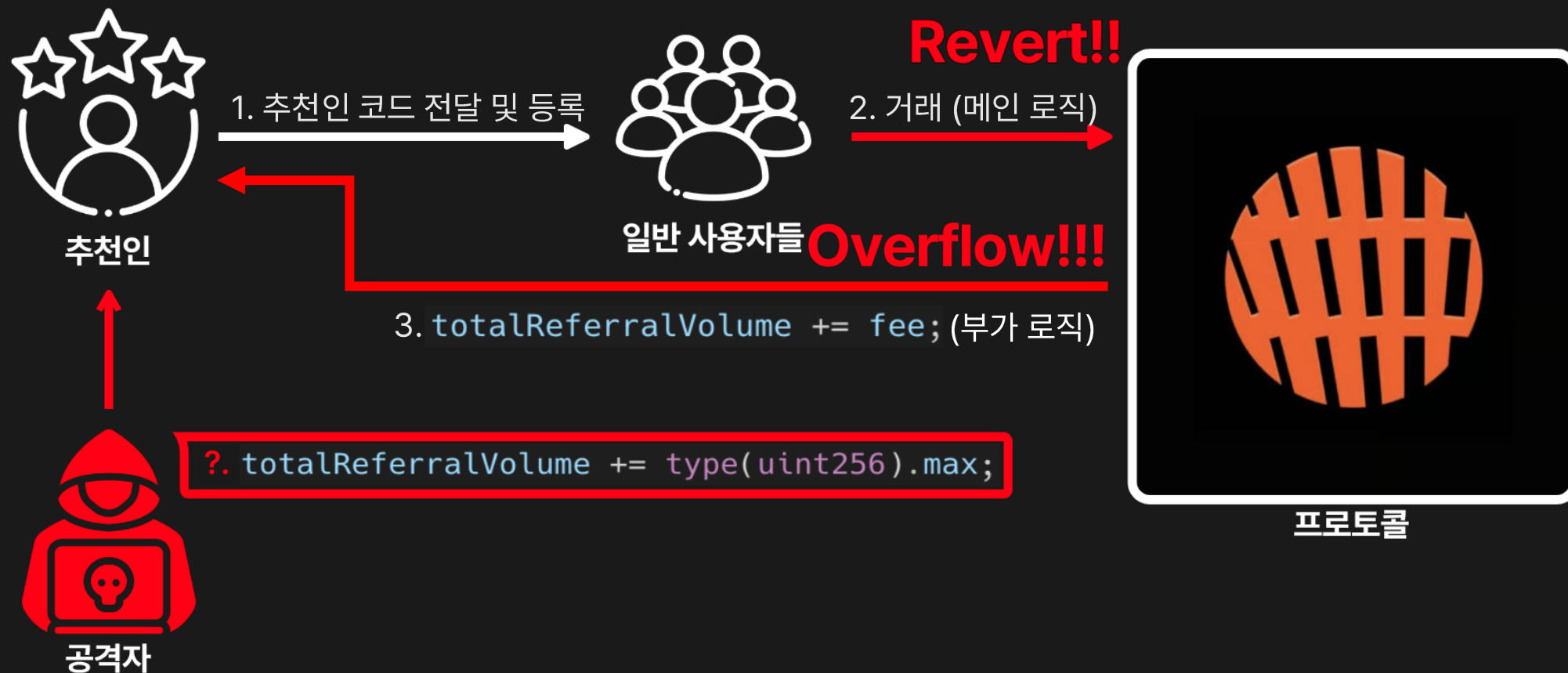
DoS via **Unrestricted Shared State** Manipulation

거래 로직이 공유 데이터에 의존하는 경우, 해당 데이터가 조작되면 전체 거래 중단 위험



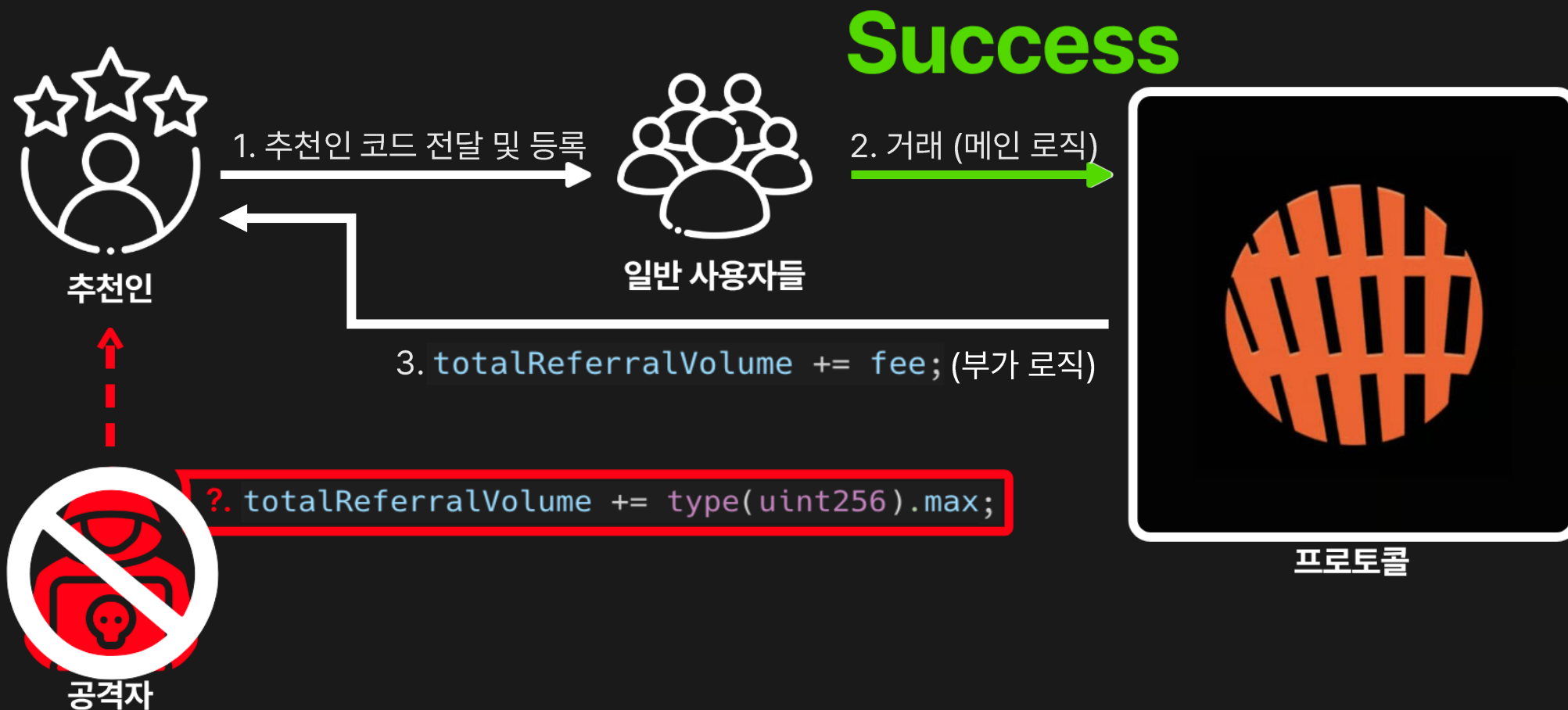
DoS via **Unrestricted Shared State** Manipulation

거래 로직이 공유 데이터에 의존하는 경우, 해당 데이터가 조작되면 전체 거래 중단 위험



Mitigation : Add Modifier

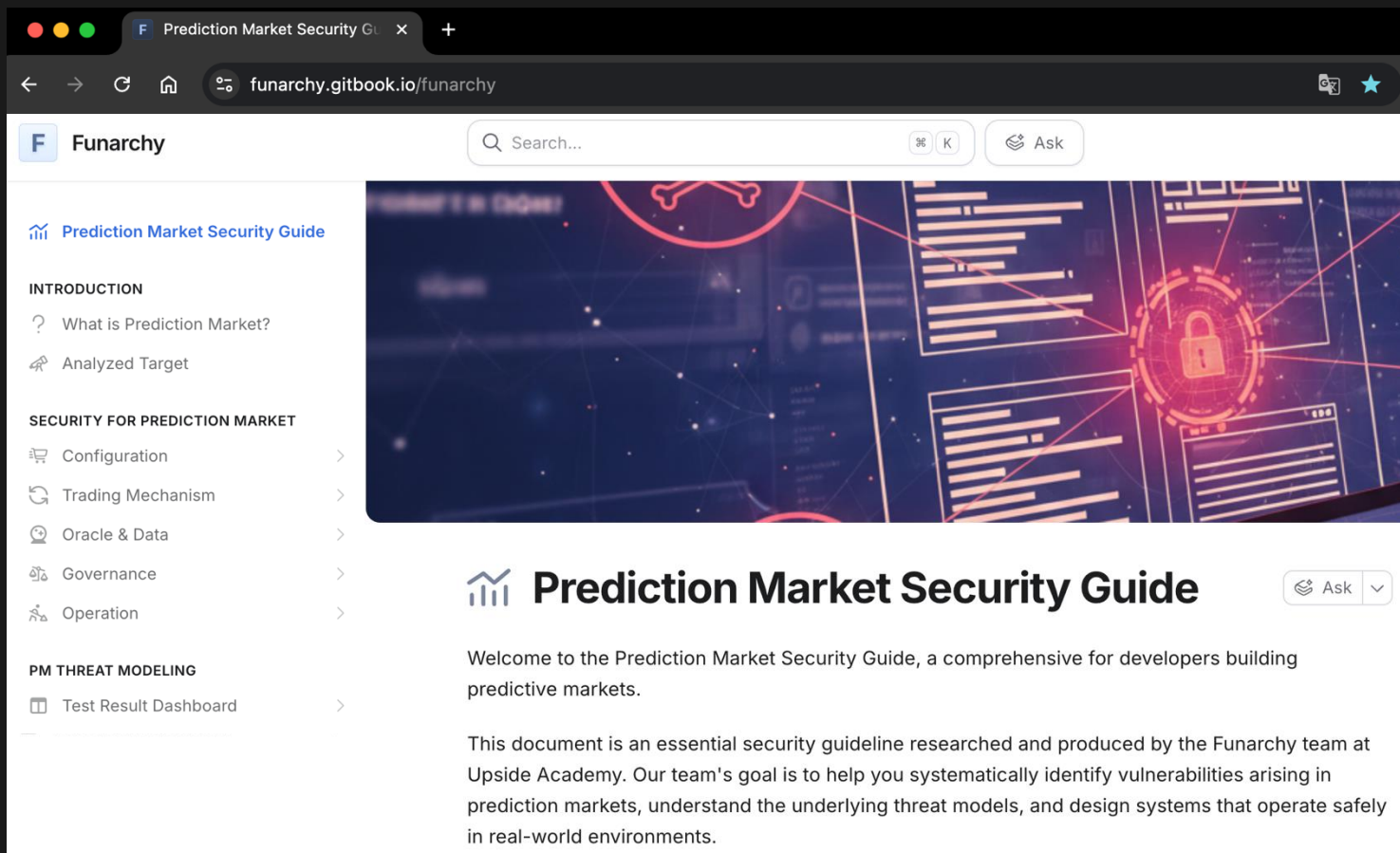
공유 데이터에 대한 접근 제한



Project Achievements

Threat Modeling Applied to Real Protocols

정의한 위협 모델을 실제 프로토콜에 적용해 구조적 취약점을 분석한 결과



The screenshot shows a web browser displaying the 'Prediction Market Security Guide' on the Funarchy website. The browser's address bar shows 'funarchy.gitbook.io/funarchy'. The page has a dark theme with a sidebar on the left containing a navigation menu. The main content area features a large header image with a padlock and circuitry, followed by the title 'Prediction Market Security Guide' and a welcome message.

Funarchy

Search...

Prediction Market Security Guide

INTRODUCTION

- ? What is Prediction Market?
- 🔍 Analyzed Target

SECURITY FOR PREDICTION MARKET

- ⚙️ Configuration
- 🔄 Trading Mechanism
- 🗉 Oracle & Data
- 🏛️ Governance
- ⚙️ Operation

PM THREAT MODELING

- 📊 Test Result Dashboard

Prediction Market Security Guide

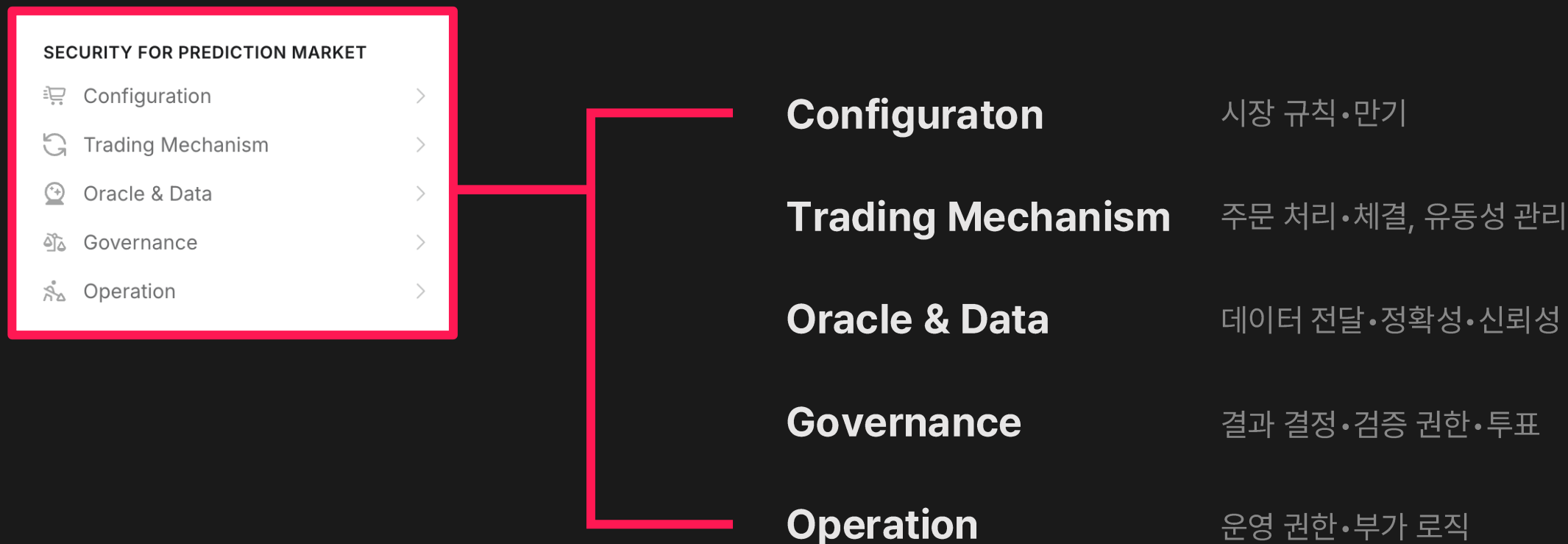
Welcome to the Prediction Market Security Guide, a comprehensive for developers building predictive markets.

This document is an essential security guideline researched and produced by the Funarchy team at Upside Academy. Our team's goal is to help you systematically identify vulnerabilities arising in prediction markets, understand the underlying threat models, and design systems that operate safely in real-world environments.



Threat Modeling Applied to Real Protocols

정의한 위협 모델을 실제 프로토콜에 적용해 구조적 취약점을 분석한 결과



Threat Modeling Applied to Real Protocols

정의한 위협 모델을 실제 프로토콜에 적용해 구조적 취약점을 분석한 결과

PM THREAT MODELING

Test Result Dashboard

Polymarket Test Results

Myriad Test Result

Seer Test Result

Rain Test Result

Overtime Test result

SK Bet Test Result

Truemarket Test Result

Azuro Test Result

Opinion Test Result

Polymarket

Configuration

Trading Mechanism

Oracle & Data

Governance

Operation

Myriad

Configuration

Trading Mechanism

Oracle & Data

Governance

Operation

Seer

Configuration

Trading Mechanism

Oracle & Data

Governance

Operation

9개 프로토콜에 대한 대시보드

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Threat Modeling Applied to Real Protocols

정의한 위협 모델을 실제 프로토콜에 적용해 구조적 취약점을 분석한 결과

PM THREAT MODELING

 **Test Result Dashboard** 

Polymarket Test Results

Myriad Test Result

Seer Test Result

Rain Test Result

Overtime Test result

SK Bet Test Result

Truemarket Test Result

Azuro Test Result

Opinion Test Result

Analysis stage

Configuration

- ☒ Ambiguous Resolution Criteria
- ☒ Manual Oracle Delay
- ☒ Extend market resolution time

Trading Mechanism

- ☒ Signature Replay Attacks
- ☒ Matching Engine Logic Errors
- ☒ State Inconsistency
 - Slippage & Sandwich Attacks
 - Permanent Loss
 - Fee bypass

Oracle & Data

Governance

Operation

Risk analysis

Oracle & Data

Single Data Source Dependency

Rules

This market will resolve to "Up" if the Bitcoin price at the end of the time range specified in the title is greater than or equal to the price at the beginning of that range. Otherwise, it will resolve to "Down".
The resolution source for this market is information from Chainlink, specifically the BTC/USD data stream available at <https://data.chainlink.streams/btc-usd>.
Please note that this market is about the price according to Chainlink data stream BTC/USD, not according to other sources or spot markets.

Created At: Dec 2, 2025, 2:26 PM

Resolution Source
<https://data.chainlink.streams/btc-usd>

- Currently, Polymarket is retrieving 15-minute market results data through a single API.

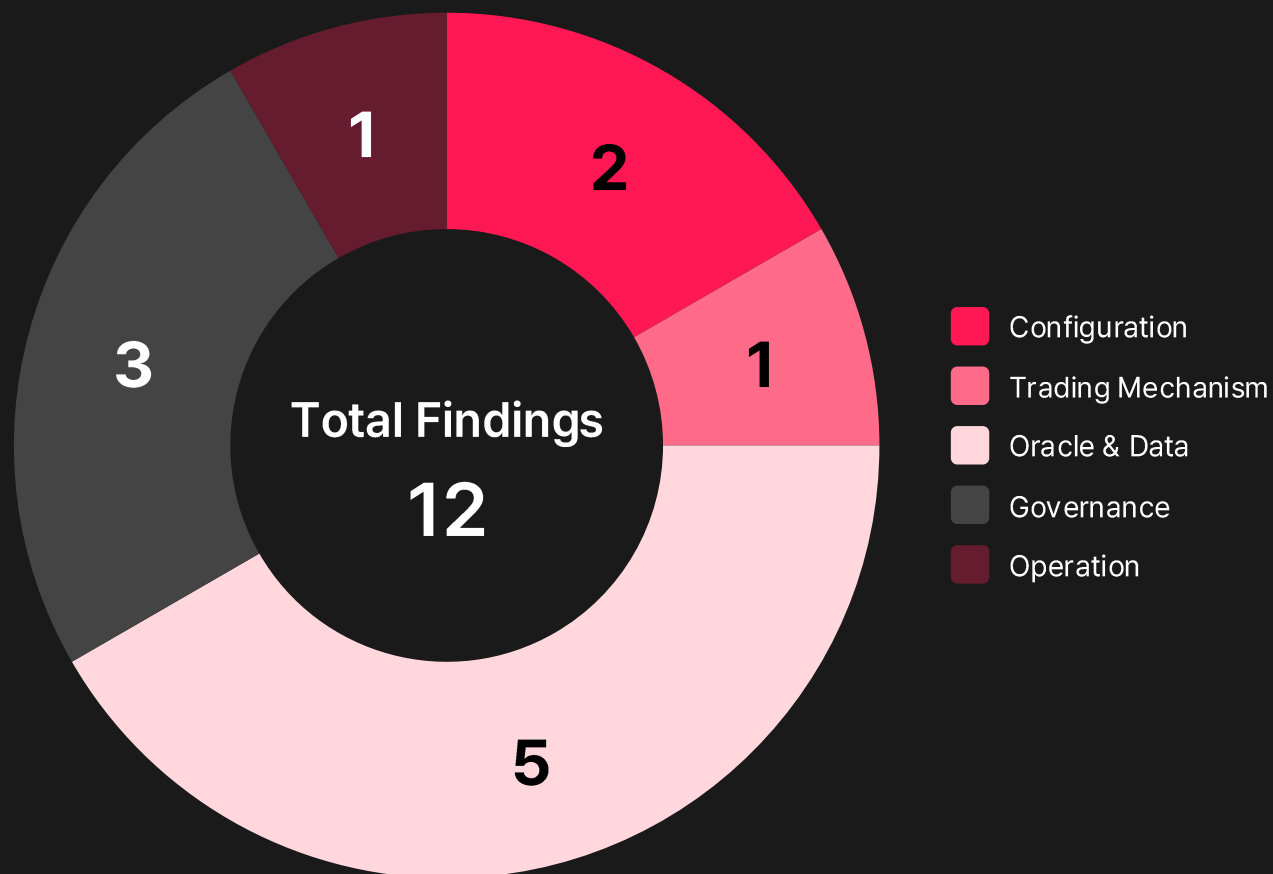
The root cause of the single API dependency problem is that the data source, which is the core of market conclusions, is tied to a centralized single point of failure. This structure results in the immediate collapse of the conclusion data in the event of a server downtime or API failure. This data collapse triggers the UMA dispute resolution system, resulting in prolonged delays in conclusions.

This results in the funds of users who entered the market expecting rapid capital turnover in the 15-minute market being frozen for a long period of time.

상세 분석 리포트

Threat Modeling Applied to Real Protocols

정의한 위협 모델을 실제 프로토콜에 적용해 구조적 취약점을 분석한 결과



Research Outcomes



취약점 제보

예측시장에 제출된
취약점 보고서 3건



위협 모델링

위협 모델링 20건 제작 및
9개의 프로토콜에 적용 및 검증



예측시장 생태계 기여

예측시장 개발자를 위한
보안 가이드 GitBook 배포

예측시장의 안전한 발전을 위한 실질적 보안 인사이트 제공

Thank you.

